JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

3D Geometry

JEE Main Mathematics · Previous Year Questions with Answer Key

388	187	2002 – 2026	203	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	144 questions
Numerical	41 questions
Out of syllabus	203 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 144 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let the foot of perpendicular from the point $(\lambda, 2, 3)$ on the line $\frac{x-4}{1} = \frac{y-9}{2} = \frac{z-5}{1}$ be the point ($1, \mu, 2$). Then the distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$ and $\frac{x-\lambda}{2} = \frac{y-\mu}{3} = \frac{z+5}{6}$ is equal to :

(A) $\frac{12}{7}$ (B) $\frac{\sqrt{145}}{7}$ (C) $\frac{\sqrt{146}}{7}$ (D) $\frac{\sqrt{143}}{7}$

Q2. JEE Main 2026 (Online) 6th April Evening Shift

The shortest distance between the lines $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z-2}{-3}$ and $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$ is:

(A) $\frac{5\sqrt{6}}{6}$ (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Let the image of the point $\mathrm{P}(1, 6, a)$ in the line $\mathrm{L} : \frac{x}{1} = \frac{y-1}{2} = \frac{z-a+1}{b}, b > 0$, be $\left(\frac{a}{3}, 0, a+c\right)$. If $\mathrm{S}(\alpha, \beta, \gamma), \alpha > 0$, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha + \beta + \gamma$ is equal to:

(A) 19 (B) 20 (C) 21 (D) 22

Q4. JEE Main 2026 (Online) 6th April Morning Shift

Let a line L be perpendicular to both the lines $\mathrm{L}_1 : \frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\mathrm{L}_2 : \frac{x-2}{1} = \frac{y-4}{4} = \frac{z-6}{7}$.

If θ is the acute angle between the lines L and $\mathrm{L}_3 : \frac{x-\frac{8}{7}}{2} = \frac{y-\frac{4}{7}}{1} = \frac{z}{2}$, then $\tan\theta$ is equal to:

(A) $\frac{3}{2}\sqrt{2}$ (B) $\frac{5}{2}\sqrt{2}$ (C) $\frac{5}{3}\sqrt{2}$ (D) $\frac{4}{3}\sqrt{2}$

Q5. JEE Main 2026 (Online) 5th April Evening Shift

Let a triangle PQR be such that P and Q lie on the line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ and are at a distance of 6 units from $R(1, 2, 3)$. If (α, β, γ) is the centroid of ΔPQR, then $\alpha + \beta + \gamma$ is equal to :

(A) 4 (B) 5 (C) 6 (D) 8

Q6. JEE Main 2026 (Online) 5th April Evening Shift

If the distance of the point $(a, 2, 5)$ from the image of the point $(1, 2, 7)$ in the line $\frac{x}{1} = \frac{y-1}{1} = \frac{z-2}{2}$ is 4 , then the sum of all possible values of a is equal to :

(A) 11 (B) 9 (C) 6 (D) 4

Q7. JEE Main 2026 (Online) 5th April Morning Shift

The square of the distance of the point $\mathrm{P}(5, 6, 7)$ from the line $\frac{x-2}{2} = \frac{y-5}{3} = \frac{z-2}{4}$ is equal to:

(A) 3 (B) 5 (C) 6 (D) 8

Q8. JEE Main 2026 (Online) 5th April Morning Shift

The square of the distance of the point of intersection of the lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(a\hat{i} - \hat{j}), a \neq 0$ and $\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + a\hat{k})$ from the origin is :

(A) 5 (B) 10 (C) 17 (D) 26

Q9. JEE Main 2026 (Online) 4th April Evening Shift

The shortest distance between the lines

$\vec{r} = \left(\frac{1}{3}\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + \frac{8}{3}\hat{\mathrm{k}}\right) + \lambda(2\hat{\mathrm{i}} - 5\hat{\mathrm{j}} + 6\hat{\mathrm{k}})$

and $\vec{r} = \left(-\frac{2}{3}\hat{\mathrm{i}} - \frac{1}{3}\hat{\mathrm{k}}\right) + \mu(\hat{\mathrm{j}} - \hat{\mathrm{k}}), \lambda, \mu \in \mathbb{R}$, is:

(A) $\sqrt{5}$ (B) 3 (C) $2\sqrt{3}$ (D) $\sqrt{15}$

Q10. JEE Main 2026 (Online) 4th April Evening Shift

If $\left(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha-1}{2}\right)$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is (are)

(A) Only 3 (B) Only 3 and - 1

(C) Only $3, \frac{1}{4}$ and -1 (D) Only 3 and $\frac{1}{4}$

Q11. JEE Main 2026 (Online) 4th April Morning Shift

A line with direction ratios $1, -1, 2$ intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q , respectively. If the length of the line segment PQ is α, then $225\alpha^2$ is equal to:

(A) 1024 (B) 1014 (C) 1104 (D) 1204

Q12. JEE Main 2026 (Online) 4th April Morning Shift

The square of the distance of the point $(-2, -8, 6)$ from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:

(A) 3 (B) 6 (C) 8 (D) 12

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

Let the point A be the foot of perpendicular drawn from the point P$(a, b, 0)$ on the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-\alpha}{3}$.

If the midpoint of the line segment PA is $\left(0, \frac{3}{4}, -\frac{1}{4}\right)$, then the value of $a^2 + b^2 + \alpha^2$ is equal to :

(A) 1 (B) 2 (C) 6 (D) 9

Q14. JEE Main 2026 (Online) 2nd April Morning Shift

If the point of intersection of the lines $\frac{x+1}{3}=\frac{y+a}{5}=\frac{z+b+1}{7}$ and $\frac{x-2}{1}=\frac{y-b}{4}=\frac{z-2a}{7}$ lies on xy-plane, then the value of $a+b$ is:

(A) 2 (B) 5 (C) 7 (D) 9

Q15. JEE Main 2026 (Online) 2nd April Morning Shift

Let a line L passing through the point (1, 1, 1) be perpendicular to both the vectors $2\hat{i}+2\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+2\hat{k}$. If $P(a,b,c)$ is the foot of perpendicular from the origin on the line L, then the value of $34(a+b+c)$ is :

(A) 50 (B) 80 (C) 100 (D) 120

Q16. JEE Main 2026 (Online) 28th January Evening Shift

Let Q(a, b, c) be the image of the point P(3, 2, 1) in the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3}=\frac{y-9}{2}=\frac{z-5}{-2}$ is

(A) 8 (B) 7 (C) 6 (D) 5

Q17. JEE Main 2026 (Online) 28th January Morning Shift

If the distances of the point $(1,2,a)$ from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines $\mathrm{L}_1:\frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and $\mathrm{L}_2:\frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal, then $a+b+c$ is equal to

(A) 4 (B) 6 (C) 7 (D) 5

Q18. JEE Main 2026 (Online) 24th January Evening Shift

The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha}=\frac{y-2}{-1}=\frac{z-4}{-\alpha}$ and $\frac{x}{\alpha}=\frac{y-1}{2}=\frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

(A) -6 (B) -8 (C) 8 (D) 6

Q19. JEE Main 2026 (Online) 24th January Morning Shift

Let the lines $\mathrm{L}_1:\vec{r}=\hat{\mathrm{i}}+2\hat{\mathrm{j}}+3\hat{\mathrm{k}}+\lambda(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+4\hat{\mathrm{k}}),\lambda\in\mathbb{R}$ and $\mathrm{L}_2:\vec{r}=(4\hat{\mathrm{i}}+\hat{\mathrm{j}})+\mu(5\hat{\mathrm{i}}+2\hat{\mathrm{j}}+\hat{\mathrm{k}}),\mu\in\mathbb{R}$, intersect at the point R . Let P and Q be the points lying on lines L_1 and L_2, respectively, such that $|\overrightarrow{\mathrm{PR}}|=\sqrt{29}$ and $|\overrightarrow{\mathrm{PQ}}|=\sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(\mathrm{QR})^2$ is equal to

(A) 360 (B) 348 (C) 320 (D) 340

Q20. JEE Main 2026 (Online) 23rd January Morning Shift

Let the direction cosines of two lines satisfy the equations : $4l + m - n = 0$ and $2mn + 10nl + 3lm = 0$.

Then the cosine of the acute angle between these lines is :

(A) $\frac{10}{7\sqrt{38}}$ (B) $\frac{10}{\sqrt{38}}$ (C) $\frac{10}{3\sqrt{38}}$ (D) $\frac{20}{3\sqrt{38}}$

Q21. JEE Main 2026 (Online) 23rd January Morning Shift

The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are $(1, 6, 3)$ and $(4, 9, \alpha)$ respectively and C is at a distance of 10 units from B. The area (in sq. units) of $\triangle ABC$ is :

(A) $20\sqrt{13}$ (B) $5\sqrt{13}$ (C) $15\sqrt{13}$ (D) $10\sqrt{13}$

Q22. JEE Main 2026 (Online) 22nd January Evening Shift

Let L be the line $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z+3}{6}$ and let S be the set of all points $(\mathrm{a}, \mathrm{b}, \mathrm{c})$ on L , whose distance from the line $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z-9}{0}$ along the line L is 7 . Then $\sum_{(a,b,c)\in S} (a+b+c)$ is equal to :

(A) 28 (B) 6 (C) 40 (D) 34

Q23. JEE Main 2026 (Online) 22nd January Morning Shift

Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance $4\sqrt{14}$ from the point $(1, -1, 0)$ and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1} = \frac{y-\beta}{2} = \frac{z-\gamma}{3}$ and $\frac{x+5}{2} = \frac{y-10}{1} = \frac{z-3}{1}$, is equal to

(A) $4\sqrt{\frac{7}{5}}$ (B) $7\sqrt{\frac{5}{4}}$ (C) $4\sqrt{\frac{5}{7}}$ (D) $2\sqrt{\frac{7}{4}}$

Q24. JEE Main 2026 (Online) 22nd January Morning Shift

If the image of the point $\mathrm{P}(1, 2, a)$ in the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{7-z}{2}$ is $\mathrm{Q}(5, b, \mathrm{c})$, then $a^2 + b^2 + c^2$ is equal to

(A) 298 (B) 264 (C) 293 (D) 283

Q25. JEE Main 2026 (Online) 21st January Evening Shift

Let the line L pass through the point $(-3, 5, 2)$ and make equal angles with the positive coordinate axes. If the distance of L from the point $(-2, r, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

(A) 16 (B) 12 (C) 6 (D) 10

Q26. JEE Main 2026 (Online) 21st January Evening Shift

Let the line L_1 be parallel to the vector $-3\hat{i}+2\hat{j}+4\hat{k}$ and pass through the point $(2,6,7)$, and the line L_2 be parallel to the vector $2\hat{i}+\hat{j}+3\hat{k}$ and pass through the point $(4,3,5)$. If the line L_3 is parallel to the vector $-3\hat{i}+5\hat{j}+16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $\left|\overrightarrow{CD}\right|^2$ is equal to:

(A) 290 (B) 171 (C) 89 (D) 312

Q27. JEE Main 2026 (Online) 21st January Morning Shift

Let (α,β,γ) be the co-ordinates of the foot of the perpendicular drawn from the point $(5,4,2)$ on the line $\overrightarrow{r}=(-\hat{i}+3\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}-\hat{k})$.

Then the length of the projection of the vector $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ on the vector $6\hat{i}+2\hat{j}+3\hat{k}$ is :

(A) $\frac{18}{7}$ (B) $\frac{15}{7}$ (C) 4 (D) 3

Q28. JEE Main 2025 (Online) 8th April Evening Shift

Let the values of λ for which the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-\lambda}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is $\frac{1}{\sqrt{6}}$ be λ_1 and λ_2. Then the radius of the circle passing through the points $(0,0),(\lambda_1,\lambda_2)$ and (λ_2,λ_1) is

(A) 3 (B) $\frac{5\sqrt{2}}{3}$ (C) $\frac{\sqrt{2}}{3}$ (D) 4

Q29. JEE Main 2025 (Online) 7th April Evening Shift

If the equation of the line passing through the point $\left(0,-\frac{1}{2},0\right)$ and perpendicular to the lines $\vec{r}=\lambda\left(\hat{i}+a\hat{j}+b\hat{k}\right)$ and $\vec{r}=\left(\hat{i}-\hat{j}-6\hat{k}\right)+\mu\left(-b\hat{i}+a\hat{j}+5\hat{k}\right)$ is $\frac{x-1}{-2}=\frac{y+4}{d}=\frac{z-c}{-4}$, then $a+b+c+d$ is equal to :

(A) 13 (B) 14 (C) 12 (D) 10

Q30. JEE Main 2025 (Online) 7th April Evening Shift

Consider the lines **L_1**: x - 1 = y - 2 = z and **L_2**: x - 2 = y = z - 1. Let the feet of the perpendiculars from the point P(5, 1, -3) on the lines **L_1** and **L_2** be Q and R respectively. If the area of the triangle PQR is A, then $4A^2$ is equal to :

(A) 151 (B) 147 (C) 139 (D) 143

Q31. JEE Main 2025 (Online) 7th April Morning Shift

Let the line L pass through $(1,1,1)$ and intersect the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-4}{2}=\frac{z}{1}$. Then, which of the following points lies on the line L ?

(A) $(7,15,13)$ (B) $(4,22,7)$ (C) $(10,-29,-50)$ (D) $(5,4,3)$

Q32. JEE Main 2025 (Online) 7th April Morning Shift

If the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x}{1}=\frac{y}{\alpha}=\frac{z-5}{1}$ is $\frac{5}{\sqrt{6}}$, then the sum of all possible values of α is

(A) $\frac{3}{2}$ (B) 3 (C) -3 (D) $-\frac{3}{2}$

Q33. JEE Main 2025 (Online) 4th April Evening Shift

Let A be the point of intersection of the lines $\mathrm{L}_1: \frac{x-7}{1}=\frac{y-5}{0}=\frac{z-3}{-1}$ and $\mathrm{L}_2: \frac{x-1}{3}=\frac{y+3}{4}=\frac{z+7}{5}$. Let B and C be the points on the lines L_1 and L_2 respectively such that $AB=AC=\sqrt{15}$. Then the square of the area of the triangle ABC is :

(A) 63 (B) 57 (C) 60 (D) 54

Q34. JEE Main 2025 (Online) 4th April Evening Shift

Let the values of p , for which the shortest distance between the lines $\frac{x+1}{3}=\frac{y}{4}=\frac{z}{5}$ and $\vec{\mathrm{r}}=(\mathrm{p}\hat{i}+2\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}+4\hat{k})$ is $\frac{1}{\sqrt{6}}$, be $\mathrm{a}, \mathrm{b}, (\mathrm{a}<\mathrm{b})$. Then the length of the latus rectum of the ellipse $\frac{x^2}{\mathrm{a}^2}+\frac{y^2}{\mathrm{b}^2}=1$ is :

(A) $\frac{3}{2}$ (B) 9 (C) 18 (D) $\frac{2}{3}$

Q35. JEE Main 2025 (Online) 4th April Morning Shift

Let the shortest distance between the lines $\frac{x-3}{3}=\frac{y-\alpha}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-\beta}{4}$ be $3\sqrt{30}$. Then the positive value of $5\alpha+\beta$ is

(A) 42 (B) 40 (C) 48 (D) 46

Q36. JEE Main 2025 (Online) 4th April Morning Shift

Let A and B be two distinct points on the line $L: \frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$. Both A and B are at a distance $2\sqrt{17}$ from the foot of perpendicular drawn from the point $(1,2,3)$ on the line L. If O is the origin, then $\overrightarrow{OA}\cdot\overrightarrow{OB}$ is equal to

(A) 49 (B) 21 (C) 47 (D) 62

Q37. JEE Main 2025 (Online) 3rd April Evening Shift

Each of the angles β and γ that a given line makes with the positive y - and z-axes, respectively, is half of the angle that this line makes with the positive x-axes. Then the sum of all possible values of the angle β is

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Q38. JEE Main 2025 (Online) 3rd April Evening Shift

The distance of the point $(7, 10, 11)$ from the line $\frac{x-4}{1} = \frac{y-4}{0} = \frac{z-2}{3}$ along the line $\frac{x-9}{2} = \frac{y-13}{3} = \frac{z-17}{6}$ is

(A) 16 (B) 12 (C) 18 (D) 14

Q39. JEE Main 2025 (Online) 3rd April Morning Shift

Let a line passing through the point $(4, 1, 0)$ intersect the line $\mathrm{L}_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at the point $A(\alpha, \beta, \gamma)$ and the line $\mathrm{L}_2 : x - 6 = y = -z + 4$ at the point $B(a, b, c)$. Then $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ is equal to

(A) 16 (B) 6 (C) 8 (D) 12

Q40. JEE Main 2025 (Online) 3rd April Morning Shift

Line L_1 passes through the point $(1, 2, 3)$ and is parallel to z-axis. Line L_2 passes through the point $(\lambda, 5, 6)$ and is parallel to y-axis. Let for $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, the shortest distance between the two lines be 3 . Then the square of the distance of the point $(\lambda_1, \lambda_2, 7)$ from the line L_1 is

(A) 25 (B) 32 (C) 40 (D) 37

Q41. JEE Main 2025 (Online) 2nd April Evening Shift

If the image of the point $\mathrm{P}(1, 0, 3)$ in the line joining the points $\mathrm{A}(4, 7, 1)$ and $\mathrm{B}(3, 5, 3)$ is $Q(\alpha, \beta, \gamma)$, then $\alpha + \beta + \gamma$ is equal to :

(A) $\frac{46}{3}$ (B) 18 (C) 13 (D) $\frac{47}{3}$

Q42. JEE Main 2025 (Online) 2nd April Evening Shift

The line L_1 is parallel to the vector $\vec{\mathrm{a}} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through the point $(7, 6, 2)$ and the line L_2 is parallel to the vector $\vec{\mathrm{b}} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through the point $(5, 3, 4)$. The shortest distance between the lines L_1 and L_2 is :

(A) $\frac{23}{\sqrt{38}}$ (B) $\frac{21}{\sqrt{38}}$ (C) $\frac{23}{\sqrt{57}}$ (D) $\frac{21}{\sqrt{57}}$

Q43. JEE Main 2025 (Online) 2nd April Morning Shift

Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$, $\mathrm{QR} = 5$ and the coordinates of the point P be $(0, 2, 3)$. If the area of the triangle PQR is $\frac{m}{n}$ then :

(A) $2\,\mathrm{m} - 5\sqrt{21}\mathrm{n} = 0$ (B) $\mathrm{m} - 5\sqrt{21}\mathrm{n} = 0$

(C) $5\,\mathrm{m} - 21\sqrt{2}\mathrm{n} = 0$ (D) $5\,\mathrm{m} - 2\sqrt{21}\mathrm{n} = 0$

Q44. JEE Main 2025 (Online) 2nd April Morning Shift

Let $ABCD$ be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the areas of the triangles ABC, ACD and ADB be 5,6 and 7 square units respectively. Then the area (in square units) of the $\triangle BCD$ is equal to :

(A) $\sqrt{110}$ (B) 12 (C) $\sqrt{340}$ (D) $7\sqrt{3}$

Q45. JEE Main 2025 (Online) 29th January Evening Shift

Let a straight line L pass through the point $P(2,-1,3)$ and be perpendicular to the lines $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-3}{-2}$ and $\frac{x-3}{1}=\frac{y-2}{3}=\frac{z+2}{4}$. If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is:

(A) $\sqrt{10}$ (B) 2 (C) $2\sqrt{3}$ (D) 3

Q46. JEE Main 2025 (Online) 29th January Evening Shift

Let P be the foot of the perpendicular from the point $(1,2,2)$ on the line $\mathrm{L}: \frac{x-1}{1}=\frac{y+1}{-1}=\frac{z-2}{2}$. Let the line $\vec{r}=(-\hat{i}+\hat{j}-2\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k}), \lambda \in \mathbf{R}$, intersect the line L at Q . Then $2(\mathrm{PQ})^2$ is equal to :

(A) 25 (B) 27 (C) 19 (D) 29

Q47. JEE Main 2025 (Online) 29th January Morning Shift

Let $\mathrm{L}_1: \frac{x-1}{1}=\frac{y-2}{-1}=\frac{z-1}{2}$ and $\mathrm{L}_2: \frac{x+1}{-1}=\frac{y-2}{2}=\frac{z}{1}$ be two lines.

Let L_3 be a line passing through the point (α,β,γ) and be perpendicular to both L_1 and L_2. If L_3 intersects L_1, then $|5\alpha-11\beta-8\gamma|$ equals :

(A) 25 (B) 20 (C) 16 (D) 18

Q48. JEE Main 2025 (Online) 29th January Morning Shift

Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+7\hat{j}+3\hat{k}$. Let $\mathrm{L}_1: \overrightarrow{\mathrm{r}}=(-\hat{i}+2\hat{j}+\hat{k})+\lambda\vec{a}, \lambda \in \mathbf{R}$ and $\mathrm{L}_2: \overrightarrow{\mathrm{r}}=(\hat{j}+\hat{k})+\mu\vec{b}, \mu \in \mathrm{R}$ be two lines. If the line L_3 passes through the point of intersection of L_1 and L_2 and is parallel to $\vec{a}+\vec{b}$, then L_3 passes through the point :

(A) $(-1,-1,1)$ (B) $(2,8,5)$ (C) $(8,26,12)$ (D) $(5,17,4)$

Q49. JEE Main 2025 (Online) 28th January Evening Shift

The square of the distance of the point $\left(\frac{15}{7},\frac{32}{7},7\right)$ from the line $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ in the direction of the vector $\hat{i}+4\hat{j}+7\hat{k}$ is:

(A) 66 (B) 54 (C) 41 (D) 44

Q50. JEE Main 2025 (Online) 28th January Morning Shift

Let $\mathrm{A}(x, y, z)$ be a point in xy-plane, which is equidistant from three points $(0, 3, 2), (2, 0, 3)$ and $(0, 0, 1)$.

Let $\mathrm{B} = (1, 4, -1)$ and $\mathrm{C} = (2, 0, -2)$. Then among the statements

(S1) : $\triangle\mathrm{ABC}$ is an isosceles right angled triangle, and

(S2) : the area of $\triangle\mathrm{ABC}$ is $\frac{9\sqrt{2}}{2}$,

(A) both are false

(B) only (S2) is true

(C) only (S1) is true

(D) both are true

Q51. JEE Main 2025 (Online) 28th January Morning Shift

If the image of the point $(4, 4, 3)$ in the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-1}{3}$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to

(A) 12 (B) 9 (C) 7 (D) 8

Q52. JEE Main 2025 (Online) 24th January Morning Shift

Let in a $\triangle ABC$, the length of the side AC be 6 , the vertex B be $(1, 2, 3)$ and the vertices A, C lie on the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Then the area (in sq. units) of $\triangle ABC$ is:

(A) 42 (B) 17 (C) 56 (D) 21

Q53. JEE Main 2025 (Online) 24th January Morning Shift

Let the line passing through the point $(-1, 2, 1)$ and parallel to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ intersect the line $\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ at the point P. Then the distance of P from the point $Q(4, -5, 1)$ is

(A) $5\sqrt{6}$ (B) 5 (C) $5\sqrt{5}$ (D) 10

Q54. JEE Main 2025 (Online) 23rd January Evening Shift

If the square of the shortest distance between the lines $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z+3}{-3}$ and $\frac{x+1}{2} = \frac{y+3}{4} = \frac{z+5}{-5}$ is $\frac{m}{n}$, where m, n are coprime numbers, then $m + n$ is equal to :

(A) 14 (B) 6 (C) 21 (D) 9

Q55. JEE Main 2025 (Online) 23rd January Evening Shift

The distance of the line $\frac{x-2}{2} = \frac{y-6}{3} = \frac{z-3}{4}$ from the point $(1, 4, 0)$ along the line $\frac{x}{1} = \frac{y-2}{2} = \frac{z+3}{3}$ is :

(A) $\sqrt{17}$ (B) $\sqrt{13}$ (C) $\sqrt{15}$ (D) $\sqrt{14}$

Q56. JEE Main 2025 (Online) 23rd January Morning Shift

Let P be the foot of the perpendicular from the point $\mathrm{Q}(10, -3, -1)$ on the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z+1}{-2}$. Then the area of the right angled triangle PQR, where R is the point $(3, -2, 1)$, is

(A) $\sqrt{30}$ (B) $9\sqrt{15}$ (C) $3\sqrt{30}$ (D) $8\sqrt{15}$

Q57. JEE Main 2025 (Online) 22nd January Evening Shift

Let a line pass through two distinct points $P(-2,-1,3)$ and Q, and be parallel to the vector $3\hat{i}+2\hat{j}+2\hat{k}$. If the distance of the point Q from the point $\mathrm{R}(1,3,3)$ is 5 , then the square of the area of $\triangle PQR$ is equal to :

(A) 148 (B) 144 (C) 136 (D) 140

Q58. JEE Main 2025 (Online) 22nd January Evening Shift

The perpendicular distance, of the line $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z+3}{2}$ from the point $\mathrm{P}(2,-10,1)$, is :

(A) 6 (B) $4\sqrt{3}$ (C) $3\sqrt{5}$ (D) $5\sqrt{2}$

Q59. JEE Main 2025 (Online) 22nd January Morning Shift

Let $\mathrm{L}_1: \frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\mathrm{L}_2: \frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2 ?

(A) $\left(\frac{14}{3},-3,\frac{22}{3}\right)$ (B) $\left(2,3,\frac{1}{3}\right)$
(C) $\left(\frac{8}{3},-1,\frac{1}{3}\right)$ (D) $\left(-\frac{5}{3},-7,1\right)$

Q60. JEE Main 2024 (Online) 9th April Evening Shift

Consider the line L passing through the points $(1,2,3)$ and $(2,3,5)$. The distance of the point $\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ from the line L along the line $\frac{3x-11}{2}=\frac{3y-11}{1}=\frac{3z-19}{2}$ is equal to

(A) 6 (B) 3 (C) 5 (D) 4

Q61. JEE Main 2024 (Online) 9th April Morning Shift

The shortest distance between the lines $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ is:

(A) $\frac{185}{\sqrt{563}}$ (B) $\frac{187}{\sqrt{563}}$ (C) $\frac{178}{\sqrt{563}}$ (D) $\frac{179}{\sqrt{563}}$

Q62. JEE Main 2024 (Online) 9th April Morning Shift

Let the line L intersect the lines $x-2=-y=z-1, 2(x+1)=2(y-1)=z+1$ and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$. Then which of the following points lies on L ?

(A) $\left(-\frac{1}{3},1,-1\right)$ (B) $\left(-\frac{1}{3},-1,1\right)$ (C) $\left(-\frac{1}{3},-1,-1\right)$ (D) $\left(-\frac{1}{3},1,1\right)$

Q63. JEE Main 2024 (Online) 8th April Evening Shift

If the shortest distance between the lines $\frac{x-\lambda}{2}=\frac{y-4}{3}=\frac{z-3}{4}$ and $\frac{x-2}{4}=\frac{y-4}{6}=\frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is :

(A) $\frac{13}{25}$ (B) 1 (C) -1 (D) $-\frac{13}{25}$

Q64. JEE Main 2024 (Online) 8th April Morning Shift

Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy-plane is the point Q. Let $OP = \gamma$; the angle between OQ and the positive x-axis be θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is

(A) $\gamma\sqrt{1 - \sin^2\phi\cos^2\theta}$

(B) $\gamma\sqrt{1 + \cos^2\theta\sin^2\phi}$

(C) $\gamma\sqrt{1 + \cos^2\phi\sin^2\theta}$

(D) $\gamma\sqrt{1 - \sin^2\theta\cos^2\phi}$

Q65. JEE Main 2024 (Online) 8th April Morning Shift

If the shortest distance between the lines

$L_1 : \vec{r} = (2 + \lambda)\hat{i} + (1 - 3\lambda)\hat{j} + (3 + 4\lambda)\hat{k}, \quad \lambda \in \mathbb{R}$

$L_2 : \vec{r} = 2(1 + \mu)\hat{i} + 3(1 + \mu)\hat{j} + (5 + \mu)\hat{k}, \quad \mu \in \mathbb{R}$

is $\frac{m}{\sqrt{n}}$, where $\gcd(m, n) = 1$, then the value of $m + n$ equals

(A) 384 (B) 387 (C) 390 (D) 377

Q66. JEE Main 2024 (Online) 6th April Evening Shift

Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the image of the point $\mathrm{Q}(3, -3, 1)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14\,\mathrm{K}$, then K is equal to :

(A) 18 (B) 81 (C) 72 (D) 36

Q67. JEE Main 2024 (Online) 6th April Morning Shift

The shortest distance between the lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ is

(A) $8\sqrt{3}$ (B) $6\sqrt{3}$ (C) $5\sqrt{3}$ (D) $4\sqrt{3}$

Q68. JEE Main 2024 (Online) 5th April Evening Shift

Let (α, β, γ) be the image of the point $(8, 5, 7)$ in the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5}$. Then $\alpha + \beta + \gamma$ is equal to :

(A) 16 (B) 20 (C) 18 (D) 14

Q69. JEE Main 2024 (Online) 5th April Morning Shift

If the line $\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4 - z$ makes a right angle with the line $\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7}$, then $4\lambda + 9\mu$ is equal to :

(A) 4 (B) 13 (C) 5 (D) 6

Q70. JEE Main 2024 (Online) 5th April Morning Shift

Let d be the distance of the point of intersection of the lines $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ and $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ from the point $(7,8,9)$. Then d^2+6 is equal to :

(A) 75 (B) 78 (C) 72 (D) 69

Q71. JEE Main 2024 (Online) 4th April Evening Shift

Let P be the point of intersection of the lines $\frac{x-2}{1}=\frac{y-4}{5}=\frac{z-2}{1}$ and $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-3}{2}$. Then, the shortest distance of P from the line $4x=2y=z$ is

(A) $\frac{3\sqrt{14}}{7}$ (B) $\frac{5\sqrt{14}}{7}$ (C) $\frac{\sqrt{14}}{7}$ (D) $\frac{6\sqrt{14}}{7}$

Q72. JEE Main 2024 (Online) 4th April Morning Shift

Let the point, on the line passing through the points $P(1,-2,3)$ and $Q(5,-4,7)$, farther from the origin and at a distance of 9 units from the point P, be (α,β,γ). Then $\alpha^2+\beta^2+\gamma^2$ is equal to :

(A) 150 (B) 155 (C) 160 (D) 165

Q73. JEE Main 2024 (Online) 1st February Evening Shift

Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point $\mathrm{R}(1,2,3)$. If the centroid of the triangle PQR is (α,β,γ), then $\alpha^2+\beta^2+\gamma^2$ is :

(A) 18 (B) 24 (C) 26 (D) 36

Q74. JEE Main 2024 (Online) 1st February Evening Shift

If the mirror image of the point $P(3,4,9)$ in the line

$\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ is (α,β,γ), then 14 $(\alpha+\beta+\gamma)$ is :

(A) 102 (B) 138 (C) 132 (D) 108

Q75. JEE Main 2024 (Online) 1st February Morning Shift

If the shortest distance between the lines

$\frac{x-\lambda}{-2}=\frac{y-2}{1}=\frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1}=\frac{y-1}{-2}=\frac{z-2}{1}$ is 1 , then the sum of all possible values of λ is :

(A) 0 (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) $-2\sqrt{3}$

Q76. JEE Main 2024 (Online) 31st January Evening Shift

Let (α,β,γ) be the mirror image of the point $(2,3,5)$ in the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$. Then, $2\alpha+3\beta+4\gamma$ is equal to

(A) 32 (B) 31 (C) 33 (D) 34

Q77. JEE Main 2024 (Online) 31st January Evening Shift

The shortest distance, between lines L_1 and L_2, where $L_1 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+4}{2}$ and L_2 is the line, passing through the points $\mathrm{A}(-4,4,3), \mathrm{B}(-1,6,3)$ and perpendicular to the line $\frac{x-3}{-2} = \frac{y}{3} = \frac{z-1}{1}$, is

(A) $\frac{141}{\sqrt{221}}$ (B) $\frac{24}{\sqrt{117}}$ (C) $\frac{42}{\sqrt{117}}$ (D) $\frac{121}{\sqrt{221}}$

Q78. JEE Main 2024 (Online) 31st January Morning Shift

The distance of the point $Q(0,2,-2)$ from the line passing through the point $P(5,-4,3)$ and perpendicular to the lines $\vec{r} = (-3\hat{i} + 2\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 5\hat{k}), \lambda \in \mathbb{R}$ and $\vec{r} = (\hat{i} - 2\hat{j} + \hat{k}) + \mu(-\hat{i} + 3\hat{j} + 2\hat{k}), \mu \in \mathbb{R}$ is :

(A) $\sqrt{74}$ (B) $\sqrt{86}$ (C) $\sqrt{54}$ (D) $\sqrt{20}$

Q79. JEE Main 2024 (Online) 30th January Evening Shift

Let $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$,

$L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k}), \mu \in \mathbb{R}$, and $L_3 : \vec{r} = \delta(\ell\hat{i} + m\hat{j} + n\hat{k}), \delta \in \mathbb{R}$

be three lines such that L_1 is perpendicular to L_2 and L_3 is perpendicular to both L_1 and L_2. Then, the point which lies on L_3 is

(A) $(1,7,-4)$ (B) $(1,-7,4)$ (C) $(-1,7,4)$ (D) $(-1,-7,4)$

Q80. JEE Main 2024 (Online) 30th January Morning Shift

Let (α, β, γ) be the foot of perpendicular from the point $(1,2,3)$ on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. Then $19(\alpha + \beta + \gamma)$ is equal to :

(A) 99 (B) 102 (C) 101 (D) 100

Q81. JEE Main 2024 (Online) 29th January Evening Shift

Let $\mathrm{P}(3,2,3), \mathrm{Q}(4,6,2)$ and $\mathrm{R}(7,3,2)$ be the vertices of $\triangle\mathrm{PQR}$. Then, the angle $\angle\mathrm{QPR}$ is

(A) $\cos^{-1}\left(\frac{7}{18}\right)$ (B) $\frac{\pi}{6}$

(C) $\cos^{-1}\left(\frac{1}{18}\right)$ (D) $\frac{\pi}{3}$

Q82. JEE Main 2024 (Online) 29th January Morning Shift

Let PQR be a triangle with $R(-1,4,2)$. Suppose $M(2,1,2)$ is the mid point of PQ. The distance of the centroid of $\triangle\mathrm{PQR}$ from the point of intersection of the lines $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ and $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ is

(A) 69 (B) $\sqrt{99}$ (C) $\sqrt{69}$ (D) 9

Q83. JEE Main 2024 (Online) 27th January Evening Shift

Let the image of the point $(1, 0, 7)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ be the point (α, β, γ). Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

(A) $(1, -2, 1+\sqrt{2})$ (B) $(3, -4, 3+2\sqrt{2})$ (C) $(3, 4, 3-2\sqrt{2})$ (D) $(1, 2, 1-\sqrt{2})$

Q84. JEE Main 2024 (Online) 27th January Morning Shift

The distance of the point $(7, -2, 11)$ from the line

$\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ along the line $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$, is :

(A) 12 (B) 18 (C) 21 (D) 14

Q85. JEE Main 2024 (Online) 27th January Morning Shift

If the shortest distance between the lines

$\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

(A) 10 (B) 5 (C) 7 (D) 8

Q86. JEE Main 2023 (Online) 15th April Morning Shift

Let S be the set of all values of λ, for which the shortest distance between the lines

$\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8\left|\sum_{\lambda \in S} \lambda\right|$ is equal to :

(A) 306 (B) 304 (C) 308 (D) 302

Q87. JEE Main 2023 (Online) 13th April Evening Shift

The line, that is coplanar to the line $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, is :

(A) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{4}$ (B) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$

(C) $\frac{x-1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ (D) $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z-5}{5}$

Q88. JEE Main 2023 (Online) 12th April Morning Shift

Let the lines $l_1 : \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ and $l_2 : 3x + 2y + z - 2 = 0 = x - 3y + 2z - 13$ be coplanar. If the point $\mathrm{P}(a, b, c)$ on l_1 is nearest to the point $\mathrm{Q}(-4, -3, 2)$, then $|a| + |b| + |c|$ is equal to

(A) 12 (B) 14 (C) 10 (D) 8

Q89. JEE Main 2023 (Online) 10th April Morning Shift

The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is :

(A) 8 (B) 7 (C) 6 (D) 9

Q90. JEE Main 2023 (Online) 8th April Morning Shift

The shortest distance between the lines $\frac{x-4}{4}=\frac{y+2}{5}=\frac{z+3}{3}$ and $\frac{x-1}{3}=\frac{y-3}{4}=\frac{z-4}{2}$ is :

(A) $3\sqrt{6}$ (B) $6\sqrt{2}$ (C) $6\sqrt{3}$ (D) $2\sqrt{6}$

Q91. JEE Main 2023 (Online) 6th April Morning Shift

One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x, y and z axes are $3, 4$ and 5 units respectively. Let P be the vertex $(3, 4, 5)$. Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is :

(A) $\frac{12}{\sqrt{5}}$ (B) $12\sqrt{5}$ (C) $\frac{12}{5}$ (D) $\frac{12}{5\sqrt{5}}$

Q92. JEE Main 2023 (Online) 1st February Morning Shift

The shortest distance between the lines

$\frac{x-5}{1}=\frac{y-2}{2}=\frac{z-4}{-3}$ and

$\frac{x+3}{1}=\frac{y+5}{4}=\frac{z-1}{-5}$ is :

(A) $7\sqrt{3}$ (B) $5\sqrt{3}$ (C) $4\sqrt{3}$ (D) $6\sqrt{3}$

Q93. JEE Main 2023 (Online) 31st January Morning Shift

Let the shortest distance between the lines

$L: \frac{x-5}{-2}=\frac{y-\lambda}{0}=\frac{z+\lambda}{1}, \lambda \geq 0$ and

$L_1: x+1=y-1=4-z$ be $2\sqrt{6}$. If (α, β, γ) lies on L,

then which of the following is NOT possible?

(A) $\alpha+2\gamma=24$ (B) $2\alpha+\gamma=7$ (C) $\alpha-2\gamma=19$ (D) $2\alpha-\gamma=9$

Q94. JEE Main 2023 (Online) 29th January Evening Shift

The shortest distance between the lines $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$ and $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$ is :

(A) $2\sqrt{3}$ (B) $3\sqrt{3}$ (C) $4\sqrt{3}$ (D) $5\sqrt{3}$

Q95. JEE Main 2023 (Online) 25th January Evening Shift

The foot of perpendicular of the point (2, 0, 5) on the line $\frac{x+1}{2}=\frac{y-1}{5}=\frac{z+1}{-1}$ is (α, β, γ). Then, which of the following is NOT correct?

(A) $\frac{\alpha}{\beta}=-8$ (B) $\frac{\alpha\beta}{\gamma}=\frac{4}{15}$

(C) $\frac{\beta}{\gamma}=-5$ (D) $\frac{\gamma}{\alpha}=\frac{5}{8}$

Q96. JEE Main 2023 (Online) 25th January Evening Shift

The shortest distance between the lines $x+1=2y=-12z$ and $x=y+2=6z-6$ is :

(A) 3 (B) $\frac{5}{2}$ (C) $\frac{3}{2}$ (D) 2

Q97. JEE Main 2023 (Online) 25th January Morning Shift

The distance of the point P(4, 6, -2) from the line passing through the point $(-3, 2, 3)$ and parallel to a line with direction ratios 3, 3, -1 is equal to :

(A) 3 (B) $\sqrt{14}$ (C) $\sqrt{6}$ (D) $2\sqrt{3}$

Q98. JEE Main 2023 (Online) 25th January Morning Shift

Consider the lines L_1 and L_2 given by

$L_1 : \frac{x-1}{2}=\frac{y-3}{1}=\frac{z-2}{2}$

$L_2 : \frac{x-2}{1}=\frac{y-2}{2}=\frac{z-3}{3}$.

A line L_3 having direction ratios 1, -1, -2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

(A) $4\sqrt{3}$ (B) $2\sqrt{6}$ (C) 4 (D) $3\sqrt{2}$

Q99. JEE Main 2022 (Online) 27th July Evening Shift

If the length of the perpendicular drawn from the point $P(a, 4, 2)$, a > 0 on the line $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a+\sum_{i=1}^{3}\alpha_i$ is equal to :

(A) 7 (B) 8 (C) 12 (D) 14

Q100. JEE Main 2022 (Online) 26th July Morning Shift

The length of the perpendicular from the point $(1, -2, 5)$ on the line passing through $(1, 2, 4)$ and parallel to the line $x+y-z=0=x-2y+3z-5$ is :

(A) $\sqrt{\frac{21}{2}}$ (B) $\sqrt{\frac{9}{2}}$ (C) $\sqrt{\frac{73}{2}}$ (D) 1

Q101. JEE Main 2022 (Online) 25th July Evening Shift

The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is :

(A) $2\sqrt{29}$ (B) 1 (C) $\sqrt{\frac{37}{29}}$ (D) $\frac{\sqrt{29}}{2}$

Q102. JEE Main 2022 (Online) 27th June Evening Shift

The shortest distance between the lines

$\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{-1}$ and $\frac{x+3}{2} = \frac{y-6}{1} = \frac{z-5}{3}$, is :

(A) $\frac{18}{\sqrt{5}}$ (B) $\frac{22}{3\sqrt{5}}$ (C) $\frac{46}{3\sqrt{5}}$ (D) $6\sqrt{3}$

Q103. JEE Main 2022 (Online) 27th June Morning Shift

If two straight lines whose direction cosines are given by the relations $l + m - n = 0$, $3l^2 + m^2 + cnl = 0$ are parallel, then the positive value of c is :

(A) 6 (B) 4 (C) 3 (D) 2

Q104. JEE Main 2022 (Online) 26th June Morning Shift

If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}$, $z = 2$ and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is :

(A) $\cos^{-1}\left(\frac{29}{4}\right)$ (B) $\sec^{-1}\left(\frac{29}{4}\right)$

(C) $\cos^{-1}\left(\frac{2}{29}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Q105. JEE Main 2022 (Online) 24th June Evening Shift

If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible value of λ is :

(A) 16 (B) 6 (C) 12 (D) 15

Q106. JEE Main 2021 (Online) 1st September Evening Shift

The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point (2, −1, 6) is :

(A) $\sqrt{26}$ (B) $2\sqrt{5}$ (C) $2\sqrt{6}$ (D) $4\sqrt{2}$

Q107. JEE Main 2021 (Online) 27th August Evening Shift

The angle between the straight lines, whose direction cosines are given by the equations 2l + 2m − n = 0 and mn + nl + lm = 0, is :

(A) $\frac{\pi}{2}$ (B) $\pi - \cos^{-1}\left(\frac{4}{9}\right)$

(C) $\cos^{-1}\left(\frac{8}{9}\right)$ (D) $\frac{\pi}{3}$

Q108. JEE Main 2021 (Online) 22th July Evening Shift

If the shortest distance between the straight lines $3(x-1) = 6(y-2) = 2(z-1)$ and $4(x-2) = 2(y-\lambda) = (z-3), \lambda \in R$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to :

(A) 3 (B) 2 (C) 5 (D) -1

Q109. JEE Main 2021 (Online) 20th July Evening Shift

The lines x = ay − 1 = z − 2 and x = 3y − 2 = bz − 2, (ab ≠ 0) are coplanar, if :

(A) b = 1, a∈R − {0}

(B) a = 1, b∈R − {0}

(C) a = 2, b = 2

(D) a = 2, b = 3

Q110. JEE Main 2021 (Online) 16th March Evening Shift

If the foot of the perpendicular from point (4, 3, 8) on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is (3, 5, 7), then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to :

(A) $\frac{1}{\sqrt{6}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{3}}$

(D) $\sqrt{\frac{2}{3}}$

Q111. JEE Main 2021 (Online) 16th March Morning Shift

Let the position vectors of two points P and Q be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are (4, −1, 2) and (−2, 1, −2), respectively. Let lines PR and QS intersect at T. If the vector $\overrightarrow{TA}$ is perpendicular to both $\overrightarrow{PR}$ and $\overrightarrow{QS}$ and the length of vector $\overrightarrow{TA}$ is $\sqrt{5}$ units, then the modulus of a position vector of A is :

(A) $\sqrt{171}$

(B) $\sqrt{227}$

(C) $\sqrt{482}$

(D) $\sqrt{5}$

Q112. JEE Main 2021 (Online) 25th February Morning Shift

The equation of the line through the point (0, 1, 2) and perpendicular to the line

$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2}$ is :

(A) $\frac{x}{3} = \frac{y-1}{-4} = \frac{z-2}{3}$

(B) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{-3}$

(C) $\frac{x}{-3} = \frac{y-1}{4} = \frac{z-2}{3}$

(D) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{3}$

Q113. JEE Main 2021 (Online) 25th February Morning Shift

Let α be the angle between the lines whose direction cosines satisfy the equations l + m − n = 0 and $l^2 + m^2 - n^2 = 0$. Then the value of $\sin^4\alpha + \cos^4\alpha$ is :

(A) $\frac{3}{8}$

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\frac{5}{8}$

Q114. JEE Main 2021 (Online) 24th February Evening Shift

Let a, b∈R. If the mirror image of the point P(a, 6, 9) with respect to the line

$\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is (20, b, −a−9), then | a + b |, is equal to :

(A) 88

(B) 90

(C) 86

(D) 84

Q115. JEE Main 2020 (Online) 5th September Morning Slot

If (a, b, c) is the image of the point (1, 2, -3) in

the line $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$, then a + b + c is :

(A) 1 (B) 2 (C) 3 (D) -1

Q116. JEE Main 2020 (Online) 3rd September Morning Slot

The lines
$\overrightarrow{r} = \left(\hat{i} - \hat{j}\right) + l\left(2\hat{i} + \widehat{k}\right)$ and
$\overrightarrow{r} = \left(2\hat{i} - \hat{j}\right) + m\left(\hat{i} + \hat{j} + \widehat{k}\right)$

(A) do not intersect for any values of l and m

(B) intersect for all values of l and m

(C) intersect when l = 2 and m = $\frac{1}{2}$

(D) intersect when l = 1 and m = 2

Q117. JEE Main 2020 (Online) 8th January Morning Slot

The shortest distance between the lines

$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and

$\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is :

(A) 3 (B) $\frac{7}{2}\sqrt{30}$ (C) $3\sqrt{30}$ (D) $2\sqrt{30}$

Q118. JEE Main 2019 (Online) 10th April Evening Slot

The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\widehat{k}$ from the straight line passing through the point (2, 3, – 4) and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\widehat{k}$ is :

(A) 6 (B) 7 (C) $2\sqrt{13}$ (D) $4\sqrt{3}$

Q119. JEE Main 2019 (Online) 10th April Morning Slot

If the length of the perpendicular from the point (β, 0, β) ($\beta \neq 0$) to the line,
$\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1}$ is $\sqrt{\frac{3}{2}}$, then β is equal to :

(A) 2 (B) 1 (C) -2 (D) -1

Q120. JEE Main 2019 (Online) 9th April Evening Slot

The vertices B and C of a ΔABC lie on the line,

$\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that BC = 5 units.

Then the area (in sq. units) of this triangle, given that the point A(1, –1, 2), is :

(A) 6 (B) $5\sqrt{17}$ (C) $\sqrt{34}$ (D) $2\sqrt{34}$

Q121. JEE Main 2019 (Online) 8th April Evening Slot

If a point R(4, y, z) lies on the line segment joining the points P(2, –3, 4) and Q(8, 0, 10), then the distance of R from the origin is :

(A) $2\sqrt{14}$ (B) $\sqrt{53}$ (C) $2\sqrt{21}$ (D) 6

Q122. JEE Main 2019 (Online) 8th April Morning Slot

The length of the perpendicular from the point (2, –1, 4) on the straight line,

$\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1}$ is :

(A) less than 2 (B) greater than 4

(C) greater than 2 but less than 3 (D) greater than 3 but less than 4

Q123. JEE Main 2019 (Online) 9th January Evening Slot

If the lines x = ay + b, z = cy + d and x = a'z + b', y = c'z + d' are perpendicular, then :

(A) ab' + bc' + 1 = 0 (B) cc' + a + a' = 0

(C) bb' + cc' + 1 = 0 (D) aa' + c + c' = 0

Q124. JEE Main 2018 (Online) 16th April Morning Slot

If the angle between the lines, $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$

and $\frac{5-x}{-2} = \frac{7y-14}{p} = \frac{z-3}{4}$ is $\cos^{-1}\left(\frac{2}{3}\right)$, then p is equal to :

(A) $\frac{7}{2}$ (B) $\frac{2}{7}$ (C) $-\frac{7}{4}$ (D) $-\frac{4}{7}$

Q125. JEE Main 2018 (Online) 15th April Evening Slot

An angle between the lines whose direction cosines are given by the equations,
l + 3m + 5n = 0 and 5lm $-$ 2mn + 6nl = 0, is :

(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{4}\right)$

(C) $\cos^{-1}\left(\frac{1}{6}\right)$ (D) $\cos^{-1}\left(\frac{1}{8}\right)$

Q126. JEE Main 2016 (Online) 10th April Morning Slot

ABC is a triangle in a plane with vertices

A(2, 3, 5), B(−1, 3, 2) and C(λ, 5, μ).

If the median through A is equally inclined to the coordinate axes, then the value of ($\lambda^3 + \mu^3 + 5$) is :

(A) 1130 (B) 1348 (C) 676 (D) 1077

Q127. JEE Main 2016 (Online) 9th April Morning Slot

The shortest distance between the lines $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$ lies in the interval :

(A) [0, 1) (B) [1, 2) (C) (2, 3] (D) (3, 4]

Q128. JEE Main 2014 (Offline)

The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q129. JEE Main 2013 (Offline)

If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar, then k can have :

(A) any value (B) exactly one value

(C) exactly two values (D) exactly three values

Q130. AIEEE 2012

If the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then k is equal to :

(A) -1 (B) $\frac{2}{9}$ (C) $\frac{9}{2}$ (D) 0

Q131. AIEEE 2011

Statement - 1 : The point $A(1, 0, 7)$ is the mirror image of the point

$B(1, 6, 3)$ in the line : $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

Statement - 2 : The line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ bisects the line

segment joining $A(1, 0, 7)$ and $B(1, 6, 3)$

(A) Statement -1 is true, Statement -2 is true; Statement -2 is **not** a correct explanation for Statement -1.

(B) Statement -1 is true, Statement - 2 is false.

(C) Statement - 1 is false , Statement -2 is true.

(D) Statement -1 is true, Statement -2 is true; Statement -2 is a correct explanation for Statement -1.

Q132. AIEEE 2010

A line AB in three-dimensional space makes angles 45° and 120° with the positive x-axis and the positive y-axis respectively. If AB makes an acute angle θ with the positive z-axis, then θ equals :

(A) 45° (B) 60° (C) 75° (D) 30°

Q133. AIEEE 2009

The projections of a vector on the three coordinate axis are $6, -3, 2$ respectively. The direction cosines of the vector are :

(A) $\frac{6}{5}, \frac{-3}{5}, \frac{2}{5}$ (B) $\frac{6}{7}, \frac{-3}{7}, \frac{2}{7}$

(C) $\frac{-6}{7}, \frac{-3}{7}, \frac{2}{7}$ (D) $6, -3, 2$

Q134. AIEEE 2008

The line passing through the points $(5, 1, a)$ and $(3, b, 1)$ crosses the yz-plane at the point $\left(0, \frac{17}{2}, \frac{-13}{2}\right)$. Then

(A) $a = 2, b = 8$ (B) $a = 4, b = 6$ (C) $a = 6, b = 4$ (D) $a = 8, b = 2$

Q135. AIEEE 2008

If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersects at a point, then the integer k is equal to

(A) -5 (B) 5 (C) 2 (D) -2

Q136. AIEEE 2007

If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

Q137. AIEEE 2007

Let L be the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$. If L makes an angle α with the positive x-axis, then cos α equals

(A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2}$

Q138. AIEEE 2006

The two lines $x = ay + b, z = cy + d$; and $x = a'y + b', z = c'y + d'$ are perpendicular to each other if :

(A) $aa' + cc' = -1$ (B) $aa' + cc' = 1$

(C) $\frac{a}{a'} + \frac{c}{c'} = -1$ (D) $\frac{a}{a'} + \frac{c}{c'} = 1$

Q139. AIEEE 2005

The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is :

(A) 0° (B) 90° (C) 45° (D) 30°

Q140. AIEEE 2004

A line makes the same angle θ, with each of the x and z axes.

If the angle β, which it makes with y-axis, is such that $\sin^2\beta = 3\sin^2\theta$, then $\cos^2\theta$ equals :

(A) $\frac{2}{5}$ (B) $\frac{1}{5}$ (C) $\frac{3}{5}$ (D) $\frac{2}{3}$

Q141. AIEEE 2004

A line with direction cosines proportional to $2, 1, 2$ meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The co-ordinates of each of the points of intersection are given by :

(A) $(2a, 3a, 3a), (2a, a, a)$ (B) $(3a, 2a, 3a), (a, a, a)$

(C) $(3a, 2a, 3a), (a, a, 2a)$ (D) $(3a, 3a, 3a), (a, a, a)$

Q142. AIEEE 2004

If the straight lines
$x = 1 + s, y = -3 - \lambda s, z = 1 + \lambda s$ and $x = \frac{t}{2}, y = 1 + t, z = 2 - t$, with parameters s and t respectively, are co-planar, then λ equals :

(A) 0 (B) -1 (C) $-\frac{1}{2}$ (D) -2

Q143. AIEEE 2003

The two lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d'$ will be perpendicular, if and only if :

(A) $aa' + cc' + 1 = 0$

(B) $aa' + bb'cc' + 1 = 0$

(C) $aa' + bb'cc' = 0$

(D) $(a + a')(b + b') + (c + c') = 0$

Q144. AIEEE 2003

The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if :

(A) $k = 3$ or -2

(B) $k = 0$ or -1

(C) $k = 1$ or -1

(D) $k = 0$ or -3

Numerical 41 questions

Q145. JEE Main 2026 (Online) 8th April Evening Shift

Let a line L_1 pass through the origin and be perpendicular to the lines

$\mathrm{L}_2 : \overrightarrow{\mathrm{r}} = (3 + \mathrm{t})\hat{i} + (2\mathrm{t} - 1)\hat{j} + (2\mathrm{t} + 4)\hat{k}$ and

$\mathrm{L}_3 : \overrightarrow{\mathrm{r}} = (3 + 2\,\mathrm{s})\hat{i} + (3 + 2\,\mathrm{s})\hat{j} + (2 + \mathrm{s})\hat{k}, \mathrm{t}, \mathrm{s} \in \mathbf{R}$.

If $(a, b, c), a \in \mathbf{Z}$, is the point on L_3 at a distance of $\sqrt{17}$ from the point of intersection of L_1 and L_2, then $(\mathrm{a} + \mathrm{b} + \mathrm{c})^2$ is equal to ____ .

Q146. JEE Main 2026 (Online) 6th April Evening Shift

Let the image of the point $\mathrm{P}(0, -5, 0)$ in the line $\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-2}$ be the point R and the image of the point $\mathrm{Q}\left(0, \frac{-1}{2}, 0\right)$ in the line $\frac{x-1}{-1} = \frac{y+9}{4} = \frac{z+1}{1}$ be the point S . Then the square of the area of the parallelogram PQRS is ____ .

Q147. JEE Main 2026 (Online) 28th January Evening Shift

If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k}), \mu \in \mathbb{R}$ along a line with direction ratios $3, -1, 0$ is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Q148. JEE Main 2026 (Online) 24th January Morning Shift

Let a line L passing through the point $\mathrm{P}(1, 1, 1)$ be perpendicular to the lines $\frac{x-4}{4} = \frac{y-1}{1} = \frac{z-1}{1}$ and $\frac{x-17}{1} = \frac{y-71}{1} = \frac{z}{0}$. Let the line L intersect the $yz-$ plane at the point Q . Another line parallel to L and passing through the point $\mathrm{S}(1, 0, -1)$ intersects the yz-plane at the point R . Then the square of the area of the parallelogram PQRS is equal to ____ .

Q149. JEE Main 2026 (Online) 23rd January Evening Shift

If the image of the point $\mathrm{P}(a, 2, a)$ in the line $\frac{x}{2}=\frac{y+a}{1}=\frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2}=\frac{y-a}{1}=\frac{z+2b}{-5}$ is P , then $a+b$ is equal to ____.

Q150. JEE Main 2025 (Online) 8th April Evening Shift

Let the area of the triangle formed by the lines $x+2=y-1=z$, $\frac{x-3}{5}=\frac{y}{-1}=\frac{z-1}{1}$ and $\frac{x}{-3}=\frac{y-3}{3}=\frac{z-2}{1}$ be A. Then A^2 is equal to _____.

Q151. JEE Main 2025 (Online) 24th January Evening Shift

Let P be the image of the point $\mathrm{Q}(7,-2,5)$ in the line $\mathrm{L}: \frac{x-1}{2}=\frac{y+1}{3}=\frac{z}{4}$ and $\mathrm{R}(5, \mathrm{p}, \mathrm{q})$ be a point on L. Then the square of the area of $\triangle PQR$ is _____.

Q152. JEE Main 2025 (Online) 22nd January Morning Shift

Let $\mathrm{L}_1: \frac{x-1}{3}=\frac{y-1}{-1}=\frac{z+1}{0}$ and $\mathrm{L}_2: \frac{x-2}{2}=\frac{y}{0}=\frac{z+4}{\alpha}, \alpha \in \mathbf{R}$, be two lines, which intersect at the point B. If P is the foot of perpendicular from the point $A(1,1,-1)$ on L_2, then the value of $26 \alpha(\mathrm{PB})^2$ is _____ .

Q153. JEE Main 2024 (Online) 9th April Evening Shift

The square of the distance of the image of the point $(6,1,5)$ in the line $\frac{x-1}{3}=\frac{y}{2}=\frac{z-2}{4}$, from the origin is _____.

Q154. JEE Main 2024 (Online) 8th April Evening Shift

Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the image of the point $\mathrm{Q}(1,6,4)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$. Then $2\alpha+\beta+\gamma$ is equal to _____

Q155. JEE Main 2024 (Online) 6th April Evening Shift

If the shortest distance between the lines $\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ and $\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to _____.

Q156. JEE Main 2024 (Online) 6th April Morning Shift

Let P be the point $(10,-2,-1)$ and Q be the foot of the perpendicular drawn from the point $R(1,7,6)$ on the line passing through the points $(2,-5,11)$ and $(-6,7,-5)$. Then the length of the line segment PQ is equal to _____.

Q157. JEE Main 2024 (Online) 5th April Evening Shift

Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____.

Q158. JEE Main 2024 (Online) 4th April Evening Shift

Consider a line L passing through the points $\mathrm{P}(1, 2, 1)$ and $\mathrm{Q}(2, 1, -1)$. If the mirror image of the point $\mathrm{A}(2, 2, 2)$ in the line L is (α, β, γ), then $\alpha + \beta + 6\gamma$ is equal to _____.

Q159. JEE Main 2024 (Online) 1st February Morning Shift

Let the line of the shortest distance between the lines

$\mathrm{L}_1 : \overrightarrow{\mathrm{r}} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and

$\mathrm{L}_2 : \overrightarrow{\mathrm{r}} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(\hat{i} + \hat{j} - \hat{k})$

intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) is the mid point of the line segment PQ, then $2(\alpha + \beta + \gamma)$ is equal to _____.

Q160. JEE Main 2024 (Online) 31st January Evening Shift

A line passes through $A(4, -6, -2)$ and $B(16, -2, 4)$. The point $P(a, b, c)$, where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points $P(a, b, c)$ and $Q(4, -12, 3)$ is equal to _____.

Q161. JEE Main 2024 (Online) 31st January Morning Shift

Let Q and R be the feet of perpendiculars from the point $\mathrm{P}(a, a, a)$ on the lines $x = y, z = 1$ and $x = -y, z = -1$ respectively. If $\angle\mathrm{QPR}$ is a right angle, then $12a^2$ is equal to _____.

Q162. JEE Main 2024 (Online) 30th January Evening Shift

Let a line passing through the point $(-1, 2, 3)$ intersect the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ at $M(\alpha, \beta, \gamma)$ and $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ at $N(a, b, c)$. Then, the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equals _____.

Q163. JEE Main 2024 (Online) 30th January Morning Shift

If d_1 is the shortest distance between the lines $x + 1 = 2y = -12z, x = y + 2 = 6z - 6$ and d_2 is the shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}, \frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$, then the value of $\frac{32\sqrt{3}\,\mathrm{d}_1}{\mathrm{d}_2}$ is :

Q164. JEE Main 2024 (Online) 29th January Evening Shift

Let O be the origin, and M and N be the points on the lines $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ and $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\overrightarrow{OM}\cdot\overrightarrow{ON}$ is equal to _____.

Q165. JEE Main 2024 (Online) 29th January Morning Shift

A line with direction ratios $2,1,2$ meets the lines $x=y+2=z$ and $x+2=2y=2z$ respectively at the points P and Q. If the length of the perpendicular from the point $(1,2,12)$ to the line PQ is l, then l^2 is _____.

Q166. JEE Main 2024 (Online) 27th January Evening Shift

The lines $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to _____.

Q167. JEE Main 2023 (Online) 11th April Morning Shift

Let a line l pass through the origin and be perpendicular to the lines

$l_1:\vec{r}=(\hat{i}-11\hat{j}-7\hat{k})+\lambda(\hat{i}+2\hat{j}+3\hat{k}),\lambda\in\mathbb{R}$ and

$l_2:\vec{r}=(-\hat{i}+\hat{\mathrm{k}})+\mu(2\hat{i}+2\hat{j}+\hat{\mathrm{k}}),\mu\in\mathbb{R}$.

If P is the point of intersection of l and l_1, and $\mathrm{Q}(\propto,\beta,\gamma)$ is the foot of perpendicular from P on l_2, then $9(\alpha+\beta+\gamma)$ is equal to _____.

Q168. JEE Main 2023 (Online) 6th April Evening Shift

If the lines $\frac{x-1}{2}=\frac{2-y}{-3}=\frac{z-3}{\alpha}$ and $\frac{x-4}{5}=\frac{y-1}{2}=\frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is _____.

Q169. JEE Main 2023 (Online) 30th January Evening Shift

Let a line L pass through the point $P(2,3,1)$ and be parallel to the line $x+3y-2z-2=0=x-y+2z$. If the distance of L from the point $(5,3,8)$ is α, then $3\alpha^2$ is equal to :

Q170. JEE Main 2023 (Online) 29th January Morning Shift

Let the co-ordinates of one vertex of ΔABC be $A(0,2,\alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. For $\alpha\in\mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____.

Q171. JEE Main 2023 (Online) 25th January Evening Shift

If the shortest distance between the line joining the points (1, 2, 3) and (2, 3, 4), and the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to _____.

Q172. JEE Main 2023 (Online) 24th January Evening Shift

If the shortest distance between the lines $\frac{x+\sqrt{6}}{2}=\frac{y-\sqrt{6}}{3}=\frac{z-\sqrt{6}}{4}$ and $\frac{x-\lambda}{3}=\frac{y-2\sqrt{6}}{4}=\frac{z+2\sqrt{6}}{5}$ is 6, then the square of sum of all possible values of λ is :

Q173. JEE Main 2023 (Online) 24th January Morning Shift

The shortest distance between the lines $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-6}{2}$ and $\frac{x-6}{3}=\frac{1-y}{2}=\frac{z+8}{0}$ is equal to _____

Q174. JEE Main 2022 (Online) 28th July Morning Shift

Let $\mathrm{P}(-2,-1,1)$ and $\mathrm{Q}\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagonal RS are $\alpha,-1,\beta$, where both α and β are integers of minimum absolute values, then $\alpha^2+\beta^2$ is equal to _____.

Q175. JEE Main 2022 (Online) 26th July Morning Shift

Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point $P(4,2,7)$. Then the square of the area of the triangle PQR is _____.

Q176. JEE Main 2022 (Online) 30th June Morning Shift

Consider a triangle ABC whose vertices are A(0, α, α), B(α, 0, α) and C(α, α, 0), α > 0. Let D be a point moving on the line x + z $-$ 3 = 0 = y and G be the centroid of ΔABC. If the minimum length of GD is $\sqrt{\frac{57}{2}}$, then α is equal to _____.

Q177. JEE Main 2022 (Online) 28th June Evening Shift

Let the image of the point P(1, 2, 3) in the line $L:\frac{x-6}{3}=\frac{y-1}{2}=\frac{z-2}{3}$ be Q. Let R (α, β, γ) be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of 22 (α + β + γ) is equal to _____.

Q178. JEE Main 2022 (Online) 25th June Evening Shift

Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively, and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____.

Q179. JEE Main 2022 (Online) 24th June Morning Shift

Let a line having direction ratios, 1, -4, 2 intersect the lines $\frac{x-7}{3}=\frac{y-1}{-1}=\frac{z+2}{1}$ and $\frac{x}{2}=\frac{y-7}{3}=\frac{z}{1}$ at the points A and B. Then $(AB)^2$ is equal to _____.

Q180. JEE Main 2022 (Online) 24th June Morning Shift

If the shortest distance between the lines

$\overrightarrow{r}=\left(-\hat{i}+3\hat{k}\right)+\lambda\left(\hat{i}-a\hat{j}\right)$

and $\overrightarrow{r}=\left(-\hat{j}+2\hat{k}\right)+\mu\left(\hat{i}-\hat{j}+\hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____.

Q181. JEE Main 2021 (Online) 20th July Morning Shift

If the shortest distance between the lines $\overrightarrow{r_1}=\alpha\hat{i}+2\hat{j}+2\hat{k}+\lambda(\hat{i}-2\hat{j}+2\hat{k})$, $\lambda\in$ R, $\alpha>0$ and $\overrightarrow{r_2}=-4\hat{i}-\hat{k}+\mu(3\hat{i}-2\hat{j}-2\hat{k})$, $\mu\in$ R is 9, then α is equal to _____.

Q182. JEE Main 2021 (Online) 25th February Evening Shift

A line 'l' passing through origin is perpendicular to the lines

$l_1:\overrightarrow{r}=(3+t)\hat{i}+(-1+2t)\hat{j}+(4+2t)\hat{k}$

$l_2:\overrightarrow{r}=(3+2s)\hat{i}+(3+2s)\hat{j}+(2+s)\hat{k}$

If the co-ordinates of the point in the first octant on 'l_2' at a distance of $\sqrt{17}$ from the point of intersection of 'l' and 'l_1' are (a, b, c) then 18(a + b + c) is equal to _____.

Q183. JEE Main 2021 (Online) 24th February Evening Shift

Let λ be an integer. If the shortest distance between the lines

$x-\lambda=2y-1=-2z$ and $x=y+2\lambda=z-\lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____.

Q184. JEE Main 2020 (Online) 9th January Morning Slot

The projection of the line segment joining the points (1, −1, 3) and (2, −4, 11) on the line joining the points (−1, 2, 3) and (3, −2, 10) is _____.

Q185. JEE Main 2020 (Online) 7th January Evening Slot

If the foot of the perpendicular drawn from the point (1, 0, 3) on a line passing through (α, 7, 1) is $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$, then α is equal to_____.

Out of syllabus 203 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q186. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

Let the foot of perpendicular of the point $P(3, -2, -9)$ on the plane passing through the points $(-1, -2, -3), (9, 3, 4), (9, -2, 1)$ be $Q(\alpha, \beta, \gamma)$. Then the distance of Q from the origin is :

(A) $\sqrt{38}$ (B) $\sqrt{29}$ (C) $\sqrt{42}$ (D) $\sqrt{35}$

Q187. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

The plane, passing through the points $(0, -1, 2)$ and $(-1, 2, 1)$ and parallel to the line passing through $(5, 1, -7)$ and $(1, -1, -1)$, also passes through the point :

(A) $(0, 5, -2)$ (B) $(2, 0, 1)$ (C) $(1, -2, 1)$ (D) $(-2, 5, 0)$

Q188. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

Let N be the foot of perpendicular from the point $\mathrm{P}(1, -2, 3)$ on the line passing through the points $(4, 5, 8)$ and $(1, -7, 5)$. Then the distance of N from the plane $2x - 2y + z + 5 = 0$ is :

(A) 7 (B) 6 (C) 9 (D) 8

Q189. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

Let the equation of plane passing through the line of intersection of the planes $x + 2y + az = 2$ and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point $(a, -c, c)$ is $\frac{2}{\sqrt{a}}$, then $\frac{a+b}{c}$ is equal to :

(A) -2 (B) 4 (C) 2 (D) -4

Q190. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(-1, 2, 3)$ from the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 10$ parallel to the line of the shortest distance between the lines $\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k})$ is :

(A) $3\sqrt{6}$ (B) $3\sqrt{5}$ (C) $2\sqrt{6}$ (D) $2\sqrt{5}$

Q191. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let the plane P: $4x - y + z = 10$ be rotated by an angle $\frac{\pi}{2}$ about its line of intersection with the plane $x + y - z = 4$. If α is the distance of the point $(2, 3, -4)$ from the new position of the plane P, then 35α is equal to :

(A) 126 (B) 105 (C) 85 (D) 90

Q192. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

Let the line passing through the points $\mathrm{P}(2, -1, 2)$ and $\mathrm{Q}(5, 3, 4)$ meet the plane $x - y + z = 4$ at the point R. Then the distance of the point R from the plane $x + 2y + 3z + 2 = 0$ measured parallel to the line $\frac{x-7}{2} = \frac{y+3}{2} = \frac{z-2}{1}$ is equal to :

(A) $\sqrt{31}$ (B) $\sqrt{189}$ (C) $\sqrt{61}$ (D) 3

Q193. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

Let P be the plane passing through the points $(5, 3, 0), (13, 3, -2)$ and $(1, 6, 2)$. For $\alpha \in \mathbb{N}$, if the distances of the points $\mathrm{A}(3, 4, \alpha)$ and $\mathrm{B}(2, \alpha, a)$ from the plane P are 2 and 3 respectively, then the positive value of a is :

(A) 6 (B) 4 (C) 5 (D) 3

Q194. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

Let (α, β, γ) be the image of the point $\mathrm{P}(2, 3, 5)$ in the plane $2x + y - 3z = 6$. Then $\alpha + \beta + \gamma$ is equal to :

(A) 10 (B) 9 (C) 5 (D) 12

Q195. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

If equation of the plane that contains the point $(-2, 3, 5)$ and is perpendicular to each of the planes $2x + 4y + 5z = 8$ and $3x - 2y + 3z = 5$ is $\alpha x + \beta y + \gamma z + 97 = 0$ then $\alpha + \beta + \gamma =$

(A) 15 (B) 16 (C) 17 (D) 18

Q196. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let the image of the point $\mathrm{P}(1, 2, 6)$ in the plane passing through the points $\mathrm{A}(1, 2, 0), \mathrm{B}(1, 4, 1)$ and $\mathrm{C}(0, 5, 1)$ be $\mathrm{Q}(\alpha, \beta, \gamma)$. Then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to :

(A) 76 (B) 62 (C) 70 (D) 65

Q197. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let the line $\frac{x}{1} = \frac{6-y}{2} = \frac{z+8}{5}$ intersect the lines $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ and $\frac{x+3}{6} = \frac{3-y}{3} = \frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is :

(A) 3 (B) $\frac{10}{3}$ (C) 4 (D) $\frac{11}{3}$

Q198. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

Let two vertices of a triangle ABC be (2, 4, 6) and (0, −2, −5), and its centroid be (2, 1, −1). If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ), then $\alpha\beta + \beta\gamma + \gamma\alpha$ is equal to :

(A) 72 (B) 74 (C) 76 (D) 70

Q199. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q, then q and 2q are the roots of the equation :

(A) $x^2 + 18x - 72 = 0$ (B) $x^2 - 18x - 72 = 0$ (C) $x^2 + 18x + 72 = 0$ (D) $x^2 - 18x + 72 = 0$

Q200. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

For $\mathrm{a}, \mathrm{b} \in \mathbb{Z}$ and $|\mathrm{a} - \mathrm{b}| \leq 10$, let the angle between the plane $\mathrm{P} : \mathrm{ax} + y - \mathrm{z} = \mathrm{b}$ and the line $l : x - 1 = \mathrm{a} - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$. If the distance of the point $(6, -6, 4)$ from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to :

(A) 48 (B) 85 (C) 32 (D) 25

Q201. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let P be the plane passing through the line

$\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point $(2, 4, -3)$.

If the image of the point $(-1, 3, 4)$ in the plane P

is (α, β, γ) then $\alpha + \beta + \gamma$ is equal to :

(A) 10 (B) 12 (C) 9 (D) 11

Q202. JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

If the equation of the plane containing the line

$x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane

$\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$

is $ax + by + cz = 4$, then $(a - b + c)$ is equal to :

(A) 18 (B) 22 (C) 20 (D) 24

Q203. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

A plane P contains the line of intersection of the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$. If P passes through the point $(0, 2, -2)$, then the square of distance of the point $(12, 12, 18)$ from the plane P is :

(A) 310 (B) 620 (C) 1240 (D) 155

Q204. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

Let the line L pass through the point $(0, 1, 2)$, intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point $\mathrm{P}(1, -9, 2)$ from the line L is :

(A) 9 (B) $\sqrt{74}$ (C) $\sqrt{69}$ (D) $\sqrt{54}$

Q205. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

If the equation of the plane passing through the line of intersection of the planes $2x - y + z = 3, 4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$, then $a + b + c$ is equal to :

(A) 13 (B) 15 (C) 14 (D) 12

Q206. JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Let the plane P pass through the intersection of the planes $2x + 3y - z = 2$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point (−7, 1, 1), then d^2 is equal to :

(A) $\frac{250}{83}$ (B) $\frac{250}{82}$ (C) $\frac{15}{53}$ (D) $\frac{25}{83}$

Q207. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

Let the image of the point $P(2, -1, 3)$ in the plane $x + 2y - z = 0$ be Q.

Then the distance of the plane $3x + 2y + z + 29 = 0$ from the point Q is :

(A) $2\sqrt{14}$ (B) $\frac{22\sqrt{2}}{7}$ (C) $\frac{24\sqrt{2}}{7}$ (D) $3\sqrt{14}$

Q208. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

Let the plane $\mathrm{P} : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to

the line $\mathrm{L} : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P

on the y-axis is 1 , then the distance between P and L is :

(A) $\frac{6}{\sqrt{14}}$ (B) $\sqrt{14}$ (C) $\sqrt{\frac{2}{7}}$ (D) $\sqrt{\frac{7}{2}}$

Q209. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points $\mathrm{A}, \mathrm{B}, \mathrm{C}$ is $(2, \mathrm{a}, 4), \mathrm{a} \in \mathrm{N}$. If the volume of the tetrahedron OABC is 144 unit^3, then which of the following points is **NOT** on P ?

(A) $(3, 0, 4)$ (B) $(0, 6, 3)$ (C) $(0, 4, 4)$ (D) $(2, 2, 4)$

Q210. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

Let P be the plane, passing through the point $(1, -1, -5)$ and perpendicular to the line joining the points $(4, 1, -3)$ and $(2, 4, 3)$. Then the distance of P from the point $(3, -2, 2)$ is :

(A) 5 (B) 4 (C) 6 (D) 7

Q211. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

A vector $\vec{v}$ in the first octant is inclined to the x-axis at 60°, to the y-axis at 45 and to the z-axis at an acute angle. If a plane passing through the points $(\sqrt{2}, -1, 1)$ and (a, b, c), is normal to $\vec{v}$, then :

(A) $a + b + \sqrt{2}c = 1$ (B) $\sqrt{2}a + b + c = 1$ (C) $\sqrt{2}a - b + c = 1$ (D) $a + \sqrt{2}b + c = 1$

Q212. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

If a plane passes through the points $(-1, k, 0), (2, k, -1), (1, 1, 2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is :

(A) $\frac{17}{5}$ (B) $\frac{6}{13}$ (C) $\frac{13}{6}$ (D) $\frac{5}{17}$

Q213. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

The line l_1 passes through the point (2, 6, 2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is :

(A) 9 (B) 7 (C) $\frac{19}{3}$ (D) $\frac{13}{3}$

Q214. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

The plane $2x - y + z = 4$ intersects the line segment joining the points A $(a, -2, 4)$ and B $(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to :

(A) $\frac{17}{3}$ (B) $\frac{97}{3}$ (C) $\frac{16}{3}$ (D) $\frac{73}{3}$

Q215. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z = a$ is :

(A) 28 (B) 22 (C) 10 (D) 16

Q216. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel to the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to :

(A) 3 (B) 1 (C) -1 (D) 5

Q217. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

Let the plane containing the line of intersection of the planes

$P_1 : x + (\lambda + 4)y + z = 1$ and

$P_2 : 2x + y + z = 2$

pass through the points (0, 1, 0) and (1, 0, 1). Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P_2 is :

(A) $2\sqrt{6}$ (B) $3\sqrt{6}$ (C) $4\sqrt{6}$ (D) $5\sqrt{6}$

Q218. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The distance of the point (7, -3, -4) from the plane passing through the points (2, -3, 1), (-1, 1, -2) and (3, -4, 2) is :

(A) $4\sqrt{2}$ (B) 4 (C) 5 (D) $5\sqrt{2}$

Q219. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(-1, 9, -16)$ from the plane

$2x + 3y - z = 5$ measured parallel to the line

$\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is :

(A) $13\sqrt{2}$ (B) 26 (C) $20\sqrt{2}$ (D) 31

Q220. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

Let Q be the foot of perpendicular drawn from the point $P(1, 2, 3)$ to the plane $x + 2y + z = 14$. If R is a point on the plane such that $\angle PRQ = 60^\circ$, then the area of $\triangle PQR$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) 3

Q221. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

If $(2, 3, 9), (5, 2, 1), (1, \lambda, 8)$ and $(\lambda, 2, 3)$ are coplanar, then the product of all possible values of λ is:

(A) $\frac{21}{2}$ (B) $\frac{59}{8}$ (C) $\frac{57}{8}$ (D) $\frac{95}{8}$

Q222. JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

If the foot of the perpendicular from the point $\mathrm{A}(-1,4,3)$ on the plane $\mathrm{P}: 2x+\mathrm{m}y+\mathrm{n}z=4$, is $\left(-2,\frac{7}{2},\frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios $3,-1,-4$, is equal to :

(A) 1 (B) $\sqrt{26}$ (C) $2\sqrt{2}$ (D) $\sqrt{14}$

Q223. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

Let the lines

$\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ and

$\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ be coplanar

and P be the plane containing these two lines.

Then which of the following points does **NOT** lie on P?

(A) $(0,-2,-2)$ (B) $(-5,0,-1)$ (C) $(3,-1,0)$ (D) $(0,4,5)$

Q224. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

A plane P is parallel to two lines whose direction ratios are $-2,1,-3$ and $-1,2,-2$ and it contains the point $(2,2,-2)$. Let P intersect the co-ordinate axes at the points $\mathrm{A},\mathrm{B},\mathrm{C}$ making the intercepts α,β,γ. If V is the volume of the tetrahedron OABC, where O is the origin, and $\mathrm{p}=\alpha+\beta+\gamma$, then the ordered pair (V,p) is equal to :

(A) $(48,-13)$ (B) $(24,-13)$ (C) $(48,11)$ (D) $(24,-5)$

Q225. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

The foot of the perpendicular from a point on the circle $x^2+y^2=1, z=0$ to the plane $2x+3y+z=6$ lies on which one of the following curves?

(A) $(6x+5y-12)^2+4(3x+7y-8)^2=1, z=6-2x-3y$

(B) $(5x+6y-12)^2+4(3x+5y-9)^2=1, z=6-2x-3y$

(C) $(6x+5y-14)^2+9(3x+5y-7)^2=1, z=6-2x-3y$

(D) $(5x+6y-14)^2+9(3x+7y-8)^2=1, z=6-2x-3y$

Q226. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

If the line of intersection of the planes $ax+by=3$ and $ax+by+cz=0$, a >0 makes an angle 30° with the plane $y-z+2=0$, then the direction cosines of the line are :

(A) $\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0$ (B) $\frac{1}{\sqrt{2}},\pm\frac{1}{\sqrt{2}},0$

(C) $\frac{1}{\sqrt{5}},-\frac{2}{\sqrt{5}},0$ (D) $\frac{1}{2},-\frac{\sqrt{3}}{2},0$

Q227. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

If the plane P passes through the intersection of two mutually perpendicular planes $2x + ky - 5z = 1$ and $3kx - ky + z = 5, k < 3$ and intercepts a unit length on positive x-axis, then the intercept made by the plane P on the y-axis is :

(A) $\frac{1}{11}$ (B) $\frac{5}{11}$ (C) 6 (D) 7

Q228. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

A plane E is perpendicular to the two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$, and passes through the point $P(1, -1, 1)$. If the distance of the plane E from the point $Q(a, a, 2)$ is $3\sqrt{2}$, then $(PQ)^2$ is equal to :

(A) 9 (B) 12 (C) 21 (D) 33

Q229. JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Let P be the plane containing the straight line $\frac{x-3}{9} = \frac{y+4}{-1} = \frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2} = \frac{y}{3} = \frac{z}{5}$ and $\frac{x}{3} = \frac{y}{7} = \frac{z}{8}$. If d is the distance of P from the point $(2, -5, 11)$, then d^2 is equal to :

(A) $\frac{147}{2}$ (B) 96 (C) $\frac{32}{3}$ (D) 54

Q230. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

The distance of the point (3, 2, -1) from the plane $3x - y + 4z + 1 = 0$ along the line $\frac{2-x}{2} = \frac{y-3}{2} = \frac{z+1}{1}$ is equal to :

(A) 9 (B) 6 (C) 3 (D) 2

Q231. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let $\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$, for some p, q $\in$ R. The shortest distance of the plane from the origin is :

(A) $\sqrt{\frac{3}{109}}$ (B) $\sqrt{\frac{5}{142}}$ (C) $\frac{5}{\sqrt{71}}$ (D) $\frac{1}{\sqrt{142}}$

Q232. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let Q be the mirror image of the point P(1, 2, 1) with respect to the plane x + 2y + 2z = 16. Let T be a plane passing through the point Q and contains the line $\vec{r} = -\hat{k} + \lambda\left(\hat{i} + \hat{j} + 2\hat{k}\right)$, $\lambda \in R$. Then, which of the following points lies on T?

(A) (2, 1, 0) (B) (1, 2, 1) (C) (1, 2, 2) (D) (1, 3, 2)

Q233. JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

If the mirror image of the point (2, 4, 7) in the plane 3x $-$ y + 4z = 2 is (a, b, c), then 2a + b + 2c is equal to :

(A) 54 (B) 50 (C) -6 (D) -42

Q234. JEE Main 2022 (Online) 28th June Evening Shift · MCQ (Single Correct Answer)

Let the plane ax + by + cz = d pass through (2, 3, −5) and is perpendicular to the planes 2x + y − 5z = 10 and 3x + 5y − 7z = 12. If a, b, c, d are integers d > 0 and gcd (|a|, |b|, |c|, d) = 1, then the value of a + 7b + c + 20d is equal to :

(A) 18 (B) 20 (C) 24 (D) 22

Q235. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

If two distinct point Q, R lie on the line of intersection of the planes $-x+2y-z=0$ and $3x-5y+2z=0$ and $PQ=PR=\sqrt{18}$ where the point P is (1, −2, 3), then the area of the triangle PQR is equal to :

(A) $\frac{2}{3}\sqrt{38}$ (B) $\frac{4}{3}\sqrt{38}$ (C) $\frac{8}{3}\sqrt{38}$ (D) $\sqrt{\frac{152}{3}}$

Q236. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x+8y+13z-29=0$ and $8x-7y+z-20=0$ and the points (2, 1, 3) and (0, 1, 2), respectively, is :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Q237. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Let the plane $P:\vec{r}.\vec{a}=d$ contain the line of intersection of two planes $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=6$ and $\vec{r}.\left(-6\hat{i}+5\hat{j}-\hat{k}\right)=7$. If the plane P passes through the point $\left(2,3,\frac{1}{2}\right)$, then the value of $\frac{|13\vec{a}|^2}{d^2}$ is equal to :

(A) 90 (B) 93 (C) 95 (D) 97

Q238. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Let the foot of the perpendicular from the point (1, 2, 4) on the line $\frac{x+2}{4}=\frac{y-1}{2}=\frac{z+1}{3}$ be P. Then the distance of P from the plane $3x+4y+12z+23=0$ is :

(A) 5 (B) $\frac{50}{13}$ (C) 4 (D) $\frac{63}{13}$

Q239. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

If the plane $2x+y-5z=0$ is rotated about its line of intersection with the plane $3x-y+4z-7=0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes through the point :

(A) (2, −2, 0) (B) (−2, 2, 0) (C) (1, 0, 2) (D) (−1, 0, −2)

Q240. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

If the lines $\vec{r} = \left(\hat{i} - \hat{j} + \hat{k}\right) + \lambda\left(3\hat{j} - \hat{k}\right)$ and $\vec{r} = \left(\alpha\hat{i} - \hat{j}\right) + \mu\left(2\hat{i} - 3\hat{k}\right)$ are co-planar, then the distance of the plane containing these two lines from the point $(\alpha, 0, 0)$ is :

(A) $\frac{2}{9}$ (B) $\frac{2}{11}$ (C) $\frac{4}{11}$ (D) 2

Q241. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let the plane 2x + 3y + z + 20 = 0 be rotated through a right angle about its line of intersection with the plane $x - 3y + 5z = 8$. If the mirror image of the point $\left(2, -\frac{1}{2}, 2\right)$ in the rotated plane is B(a, b, c), then :

(A) $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$ (B) $\frac{a}{4} = \frac{b}{5} = \frac{c}{-2}$

(C) $\frac{a}{8} = \frac{b}{-5} = \frac{c}{4}$ (D) $\frac{a}{4} = \frac{b}{5} = \frac{c}{2}$

Q242. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

Let p be the plane passing through the intersection of the planes $\vec{r}.\left(\hat{i} + 3\hat{j} - \hat{k}\right) = 5$ and $\vec{r}.\left(2\hat{i} - \hat{j} + \hat{k}\right) = 3$, and the point (2, 1, −2). Let the position vectors of the points X and Y be $\hat{i} - 2\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 2\hat{k}$ respectively. Then the points :

(A) X and X + Y are on the same side of P

(B) Y and $Y - X$ are on the opposite sides of P

(C) X and Y are on the opposite sides of P

(D) X + Y and $X - Y$ are on the same side of P

Q243. JEE Main 2022 (Online) 25th June Morning Shift · MCQ (Single Correct Answer)

Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S : x + y + z = 5. If a line L passing through $(1, -1, -1)$, parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

(A) 2 (B) 5 (C) 7 (D) 11

Q244. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

Let the points on the plane P be equidistant from the points $(-4, 2, 1)$ and $(2, -2, 3)$. Then the acute angle between the plane P and the plane 2x + y + 3z = 1 is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Q245. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

Let the acute angle bisector of the two planes $x - 2y - 2z + 1 = 0$ and $2x - 3y - 6z + 1 = 0$ be the plane P. Then which of the following points lies on P?

(A) $\left(3, 1, -\frac{1}{2}\right)$ (B) $\left(-2, 0, -\frac{1}{2}\right)$

(C) $(0, 2, -4)$ (D) $(4, 0, -2)$

Q246. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

The distance of the point $(-1, 2, -2)$ from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{5}{2}$ (C) $\frac{\sqrt{42}}{2}$ (D) $\frac{\sqrt{34}}{2}$

Q247. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the
planes $3x - 2y + 4z - 7 = 0$
and $x + 5y - 2z + 9 = 0$, be
$\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is equal to :

(A) -23 (B) -15 (C) 23 (D) 15

Q248. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

The equation of the plane passing through the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and parallel to the x-axis is :

(A) $\vec{r}.\left(\hat{j}-3\hat{k}\right)+6=0$ (B) $\vec{r}.\left(\hat{i}+3\hat{k}\right)+6=0$

(C) $\vec{r}.\left(\hat{i}-3\hat{k}\right)+6=0$ (D) $\vec{r}.\left(\hat{j}-3\hat{k}\right)-6=0$

Q249. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are $2, 3, -6$ is :

(A) 3 (B) 5 (C) 2 (D) 1

Q250. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is :

(A) $3x-y-5z+2=0$ (B) $3x-4z+3=0$

(C) $-x+2y+2z-3=0$ (D) $4x-y-5z+2=0$

Q251. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

Let P be the plane passing through the point $(1, 2, 3)$ and the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=16$ and $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$. Then which of the following points does NOT lie on P?

(A) $(3, 3, 2)$ (B) $(6, -6, 2)$ (C) $(4, 2, 2)$ (D) $(-8, 8, 6)$

Q252. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P?

(A) $(-1, 1, 2)$ (B) $(0, 1, 1)$ (C) $(1, 0, 1)$ (D) $(2, -1, 1)$

Q253. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines

$\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$ and $\frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$, lies on the plane x + 2y − z = 8, then $\alpha - \beta$ is equal to :

(A) 5 (B) 9 (C) 3 (D) 7

Q254. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Let the plane passing through the point $(-1, 0, -2)$ and perpendicular to each of the planes 2x + y − z = 2 and x − y − z = 3 be ax + by + cz + 8 = 0. Then the value of a + b + c is equal to :

(A) 3 (B) 8 (C) 5 (D) 4

Q255. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Let the foot of perpendicular from a point P(1, 2, −1) to the straight line $L : \frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ be N. Let a line be drawn from P parallel to the plane x + y + 2z = 0 which meets L at point Q. If α is the acute angle between the lines PN and PQ, then cosα is equal to _____.

(A) $\frac{1}{\sqrt{5}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2\sqrt{3}}$

Q256. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Let L be the line of intersection of planes $\vec{r}.(\hat{i} - \hat{j} + 2\hat{k}) = 2$ and $\vec{r}.(2\hat{i} + \hat{j} - \hat{k}) = 2$. If $P(\alpha, \beta, \gamma)$ is the foot of perpendicular on L from the point (1, 2, 0), then the value of $35(\alpha + \beta + \gamma)$ is equal to :

(A) 101 (B) 119 (C) 143 (D) 134

Q257. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

Consider the line L given by the equation

$\frac{x-3}{2} = \frac{y-1}{1} = \frac{z-2}{1}$.

Let Q be the mirror image of the point $(2, 3, -1)$ with respect to L. Let a plane P be such that it passes through Q, and the line L is perpendicular to P. Then which of the following points is on the plane P?

(A) $(-1, 1, 2)$ (B) $(1, 1, 1)$ (C) $(1, 1, 2)$ (D) $(1, 2, 2)$

Q258. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

If the equation of plane passing through the mirror image of a point (2, 3, 1) with respect to line $\frac{x+1}{2} = \frac{y-3}{1} = \frac{z+2}{-1}$ and containing the line $\frac{x-2}{3} = \frac{1-y}{2} = \frac{z+1}{1}$ is $\alpha x + \beta y + \gamma z = 24$, then $\alpha + \beta + \gamma$ is equal to :

(A) 21 (B) 19 (C) 18 (D) 20

Q259. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

The equation of the plane which contains the y-axis and passes through the point (1, 2, 3) is :

(A) x + 3z = 0 (B) $3x - z = 0$ (C) x + 3z = 10 (D) 3x + z = 6

Q260. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

If (x, y, z) be an arbitrary point lying on a plane P which passes through the points (42, 0, 0), (0, 42, 0) and (0, 0, 42), then the value of the expression $3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$ is equal to :

(A) 3 (B) 39 (C) -45 (D) 0

Q261. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Let P be a plane lx + my + nz = 0 containing

the line, $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$. If plane P divides the line segment AB joining

points A$(-3, -6, 1)$ and B(2, 4, -3) in ratio k : 1 then the value of k is equal to :

(A) 2 (B) 3 (C) 1.5 (D) 4

Q262. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

If for a > 0, the feet of perpendiculars from the points A(a, $-2a$, 3) and B(0, 4, 5) on the plane lx + my + nz = 0 are points C(0, $-a$, -1) and D respectively, then the length of line segment CD is equal to :

(A) $\sqrt{41}$ (B) $\sqrt{55}$ (C) $\sqrt{31}$ (D) $\sqrt{66}$

Q263. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

If the mirror image of the point (1, 3, 5) with respect to the plane

$4x - 5y + 2z = 8$ is (α, β, γ), then $5(\alpha + \beta + \gamma)$ equals :

(A) 39 (B) 41 (C) 47 (D) 43

Q264. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

Let L be a line obtained from the intersection of two planes x + 2y + z = 6 and y + 2z = 4. If point P(α, β, γ) is the foot of perpendicular from (3, 2, 1) on L, then the value of 21($\alpha + \beta + \gamma$) equals :

(A) 102 (B) 142 (C) 136 (D) 68

Q265. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

Consider the three planes

$P_1 : 3x + 15y + 21z = 9$,

$P_2 : x - 3y - z = 5$, and

$P_3 : 2x + 10y + 14z = 5$

Then, which one of the following is true?

(A) P_1 and P_2 are parallel.

(B) P_1, P_2 and P_3 all are parallel.

(C) P_1 and P_3 are parallel.

(D) P_2 and P_3 are parallel.

Q266. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

If (1, 5, 35), (7, 5, 5), (1, λ, 7) and (2λ, 1, 2) are coplanar, then the sum of all possible values of λ is :

(A) $-\frac{44}{5}$ (B) $-\frac{39}{5}$ (C) $\frac{44}{5}$ (D) $\frac{39}{5}$

Q267. JEE Main 2021 (Online) 25th February Evening Shift · MCQ (Single Correct Answer)

A plane passes through the points A(1, 2, 3), B(2, 3, 1) and C(2, 4, 2). If O is the origin and P is (2, -1, 1), then the projection of $\overrightarrow{OP}$ on this plane is of length :

(A) $\sqrt{\frac{2}{7}}$ (B) $\sqrt{\frac{2}{5}}$ (C) $\sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{2}{11}}$

Q268. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

The vector equation of the plane passing through the intersection

of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(\hat{i}-2\hat{j}\right)=-2$, and the point (1, 0, 2) is :

(A) $\vec{r}.\left(\hat{i}+7\hat{j}+3\hat{k}\right)=\frac{7}{3}$

(B) $\vec{r}.\left(\hat{i}+7\hat{j}+3\hat{k}\right)=7$

(C) $\vec{r}.\left(3\hat{i}+7\hat{j}+3\hat{k}\right)=7$

(D) $\vec{r}.\left(\hat{i}-7\hat{j}+3\hat{k}\right)=\frac{7}{3}$

Q269. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The equation of the plane passing through the point (1, 2, -3) and perpendicular to the planes

3x + y - 2z = 5 and 2x - 5y - z = 7, is :

(A) 6x - 5y + 2z + 10 =0

(B) 3x - 10y - 2z + 11 = 0

(C) 6x - 5y - 2z - 2 = 0

(D) 11x + y + 17z + 38 = 0

Q270. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The distance of the point (1, 1, 9) from the point of intersection of the line $\frac{x-3}{1}=\frac{y-4}{2}=\frac{z-5}{2}$ and the plane x + y + z = 17 is :

(A) $19\sqrt{2}$

(B) $2\sqrt{19}$

(C) 38

(D) $\sqrt{38}$

Q271. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

A plane P meets the coordinate axes at A, B and C respectively. The centroid of ΔABC is given to be (1, 1, 2). Then the equation of the line through this centroid and perpendicular to the plane P is :

(A) $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-2}{2}$

(B) $\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-2}{1}$

(C) $\frac{x-1}{2}=\frac{y-1}{2}=\frac{z-2}{1}$

(D) $\frac{x-1}{1}=\frac{y-1}{2}=\frac{z-2}{2}$

Q272. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

The shortest distance between the lines

$\frac{x-1}{0}=\frac{y+1}{-1}=\frac{z}{1}$

and x + y + z + 1 = 0, 2x − y + z + 3 = 0 is :

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\frac{1}{\sqrt{3}}$

Q273. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

If for some $\alpha \in R$, the lines

$L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$ and

$L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$ are coplanar,

then the line L_2 passes through the point :

(A) (10, 2, 2) (B) (2, −10, −2) (C) (10, −2, −2) (D) (−2, 10, 2)

Q274. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

The distance of the point (1, −2, 3) from

the plane x − y + z = 5 measured parallel to

the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is :

(A) 7 (B) 1 (C) $\frac{1}{7}$ (D) $\frac{7}{5}$

Q275. JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

The plane which bisects the line joining, the points (4, −2, 3) and (2, 4, −1) at right angles also passes through the point :

(A) (4, 0, 1) (B) (0, −1, 1) (C) (0, 1, −1) (D) (4, 0, −1)

Q276. JEE Main 2020 (Online) 3rd September Morning Slot · MCQ (Single Correct Answer)

The foot of the perpendicular drawn from the point (4, 2, 3) to the line joining the points (1, −2, 3) and (1, 1, 0) lies on the plane :

(A) x − 2y + z = 1 (B) x + 2y − z = 1 (C) x − y − 2z = 1 (D) 2x + y − z = 1

Q277. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

A plane passing through the point (3, 1, 1) contains two lines whose direction ratios are 1, −2, 2 and 2, 3, −1 respectively. If this plane also passes through the point (α, −3, 5), then α is equal to:

(A) -10 (B) 10 (C) 5 (D) -5

Q278. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

The plane passing through the points (1, 2, 1),
(2, 1, 2) and parallel to the line, 2x = 3y, z = 1
also passes through the point :

(A) (0, 6, −2) (B) (−2, 0, 1) (C) (0, −6, 2) (D) (2, 0 −1)

Q279. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

The mirror image of the point (1, 2, 3) in a plane is

$\left(-\frac{7}{3}, -\frac{4}{3}, -\frac{1}{3}\right)$. Which of the following points lies on this plane ?

(A) (1, −1, 1) (B) (−1, −1, −1) (C) (−1, −1, 1) (D) (1, 1, 1)

Q280. JEE Main 2020 (Online) 7th January Morning Slot · MCQ (Single Correct Answer)

Let P be a plane passing through the points (2, 1, 0), (4, 1, 1) and (5, 0, 1) and R be any point (2, 1, 6). Then the image of R in the plane P is :

(A) (4, 3, 2) (B) (6, 5, - 2) (C) (3, 4, -2) (D) (6, 5, 2)

Q281. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

A plane which bisects the angle between the two given planes 2x − y + 2z − 4 = 0 and x + 2y + 2z − 2 = 0, passes through the point :

(A) (1, −4, 1) (B) (1, 4, −1) (C) (2, 4, 1) (D) (2, −4, 1)

Q282. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The length of the perpendicular drawn from the point (2, 1, 4) to the plane containing the lines $\overrightarrow{r} = \left(\hat{i} + \hat{j}\right) + \lambda\left(\hat{i} + 2\hat{j} - \hat{k}\right)$ and $\overrightarrow{r} = \left(\hat{i} + \hat{j}\right) + \mu\left(-\hat{i} + \hat{j} - 2\hat{k}\right)$ is :

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) 3 (D) $\sqrt{3}$

Q283. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the plane 2x + 3y − z + 13 = 0 at a point P and the plane 3x + y + 4z = 16 at a point Q, then PQ is equal to :

(A) $2\sqrt{7}$ (B) 14 (C) $2\sqrt{14}$ (D) $\sqrt{14}$

Q284. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

If the plane 2x − y + 2z + 3 = 0 has the distances $\frac{1}{3}$ and $\frac{2}{3}$ units from the planes 4x − 2y + 4z + λ = 0 and 2x − y + 2z + μ = 0, respectively, then the maximum value of $\lambda + \mu$ is equal to :

(A) 13 (B) 9 (C) 5 (D) 15

Q285. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane x + y + z = 3 such that the foot of the perpendicular Q also lies on the plane x − y + z = 3. Then the co-ordinates of Q are :

(A) (4, 0, − 1) (B) (2, 0, 1) (C) (1, 0, 2) (D) (− 1, 0, 4)

Q286. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If Q(0, −1, −3) is the image of the point P in the plane 3x − y + 4z = 2 and R is the point (3, −1, −2), then the area (in sq. units) of ΔPQR is :

(A) $\frac{\sqrt{65}}{2}$ (B) $2\sqrt{13}$ (C) $\frac{\sqrt{91}}{2}$ (D) $\frac{\sqrt{91}}{4}$

Q287. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

Let P be the plane, which contains the line of intersection of the planes, x + y + z − 6 = 0 and 2x + 3y + z + 5 = 0 and it is perpendicular to the xy-plane. Then the distance of the point (0, 0, 256) from P is equal to :

(A) $205\sqrt{5}$ (B) $63\sqrt{5}$ (C) $11/\sqrt{5}$ (D) $17/\sqrt{5}$

Q288. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

A plane passing through the points (0, −1, 0) and (0, 0, 1) and making an angle $\frac{\pi}{4}$ with the plane y − z + 5 = 0, also passes through the point

(A) $\left(\sqrt{2}, 1, 4\right)$ (B) $\left(-\sqrt{2}, 1, 4\right)$ (C) $\left(-\sqrt{2}, -1, -4\right)$ (D) $\left(\sqrt{2}, -1, 4\right)$

Q289. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the line, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ meets the plane, x + 2y + 3z = 15 at a point P, then the distance of P from the origin is :

(A) $\frac{\sqrt{5}}{2}$ (B) $2\sqrt{5}$ (C) 9/2 (D) 7/2

Q290. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

The vector equation of the plane through the line of intersection of the planes x + y + z = 1 and 2x + 3y+ 4z = 5 which is perpendicular to the plane x − y + z = 0 is :

(A) $\overrightarrow{r} \times \left(\hat{i} - \hat{k}\right) - 2 = 0$

(B) $\overrightarrow{r}.\left(\hat{i} + \hat{k}\right) + 2 = 0$

(C) $\overrightarrow{r}.\left(\hat{i} - \hat{k}\right) + 2 = 0$

(D) $\overrightarrow{r} \times \left(\hat{i} - \hat{k}\right) + 2 = 0$

Q291. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The equation of a plane containing the line of intersection of the planes 2x − y − 4 = 0 and y + 2z − 4 = 0 and passing through the point (1, 1, 0) is :

(A) x − 3y − 2z = −2 (B) 2x − z = 2 (C) x − y − z = 0 (D) x + 3y + z = 4

Q292. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

Let S be the set of all real values of λ such that a plane passing through the points $(-\lambda^2, 1, 1)$, $(1, -\lambda^2, 1)$ and $(1, 1, -\lambda^2)$ also passes through the point $(-1, -1, 1)$. Then S is equal to :

(A) $\{1, -1\}$ (B) $\{3, -3\}$

(C) $\left\{\sqrt{3}\right\}$ (D) $\left\{\sqrt{3}, -\sqrt{3}\right\}$

Q293. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

If an angle between the line, $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z-3}{-2}$ and the plane, $x - 2y - kz = 3$ is $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$, then a value of k is :

(A) $\sqrt{\frac{3}{5}}$ (B) $-\frac{5}{2}$ (C) $-\frac{3}{2}$ (D) $\sqrt{\frac{5}{3}}$

Q294. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

The perpendicular distance from the origin to the plane containing the two lines,

$\frac{x+2}{3} = \frac{y-2}{5} = \frac{z+5}{7}$ and

$\frac{x-1}{1} = \frac{y-4}{4} = \frac{z+4}{7}$, is :

(A) $6\sqrt{11}$ (B) $\frac{11}{\sqrt{6}}$ (C) 11 (D) $11\sqrt{6}$

Q295. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

A tetrahedron has vertices P(1, 2, 1), Q(2, 1, 3), R(−1, 1, 2) and O(0, 0, 0). The angle between the faces OPQ and PQR is :

(A) $\cos^{-1}\left(\frac{17}{31}\right)$ (B) $\cos^{-1}\left(\frac{9}{35}\right)$

(C) $\cos^{-1}\left(\frac{19}{35}\right)$ (D) $\cos^{-1}\left(\frac{7}{31}\right)$

Q296. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Two lines $\frac{x-3}{1} = \frac{y+1}{3} = \frac{z-6}{-1}$ and $\frac{x+5}{7} = \frac{y-2}{-6} = \frac{z-3}{4}$ intersect at the point R. The reflection of R in the xy-plane has coordinates :

(A) (2, 4, 7) (B) (2, − 4, −7) (C) (2, − 4, 7) (D) (− 2, 4, 7)

Q297. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

If the point $(2, \alpha, \beta)$ lies on the plane which passes through the points (3, 4, 2) and (7, 0, 6) and is perpendicular to the plane 2x − 5y = 15, then $2\alpha - 3\beta$ is equal to

(A) 12 (B) 7 (C) 17 (D) 5

Q298. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The plane containing the line $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z-1}{3}$ and also containing its projection on the plane 2x + 3y − z = 5, contains which one of the following points ?

(A) (− 2, 2, 2) (B) (2, 2, 0) (C) (2, 0, − 2) (D) (0, − 2, 2)

Q299. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The direction ratios of normal to the plane through the points (0, −1, 0) and (0, 0, 1) and making an angle $\frac{\pi}{4}$ with the plane y − z + 5 = 0 are :

(A) 2, −1, 1 (B) $2\sqrt{3}, 1, -1$ (C) $\sqrt{2}, 1, -1$ (D) $\sqrt{2}, -\sqrt{2}$

Q300. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

On which of the following lines lies the point of intersection of the line, $\frac{x-4}{2} = \frac{y-5}{2} = \frac{z-3}{1}$ and the plane, x + y + z = 2 ?

(A) $\frac{x-4}{1} = \frac{y-5}{1} = \frac{z-5}{-1}$ (B) $\frac{x-2}{2} = \frac{y-3}{2} = \frac{z+3}{3}$

(C) $\frac{x-1}{1} = \frac{y-3}{2} = \frac{z+4}{-5}$ (D) $\frac{x+3}{3} = \frac{4-y}{3} = \frac{z+1}{-2}$

Q301. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

The plane which bisects the line segment joining the points (−3, −3, 4) and (3, 7, 6) at right angles, passes through which one of the following points ?

(A) (2, 1, 3) (B) (4, − 1, 2) (C) (4, 1, − 2) (D) (− 2, 3, 5)

Q302. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

The plane passing through the point (4, −1, 2) and parallel to the lines $\frac{x+2}{3} = \frac{y-2}{-1} = \frac{z+1}{2}$ and $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$ also passes through the point -

(A) (1, 1, − 1) (B) (1, 1, 1) (C) (− 1, − 1, −1) (D) (− 1, − 1, 1)

Q303. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Let A be a point on the line $\vec{r} = (1-3\mu)\hat{i} + (\mu-1)\hat{j} + (2+5\mu)\hat{k}$ and B(3, 2, 6) be a point in the space. Then the value of μ for which the vector $\overrightarrow{AB}$ is parallel to the plane x − 4y + 3z = 1 is -

(A) $\frac{1}{8}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{4}$

Q304. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ is :

(A) x − 2y + z = 0 (B) 3x + 2y − 3z = 0 (C) x + 2y − 2z = 0 (D) 5x + 2y − 4z = 0

Q305. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

The equation of the line passing through (−4, 3, 1), parallel

to the plane x + 2y − z − 5 = 0 and intersecting

the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$ is :

(A) $\frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$

(B) $\frac{x+4}{1} = \frac{y-3}{1} = \frac{z-1}{3}$

(C) $\frac{x+4}{-1} = \frac{y-3}{1} = \frac{z-1}{1}$

(D) $\frac{x-4}{2} = \frac{y+3}{1} = \frac{z+1}{4}$

Q306. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

The plane through the intersection of the planes x + y + z = 1 and 2x + 3y − z + 4 = 0 and parallel to y-axis also passes through the point :

(A) (−3, 0, -1)

(B) (3, 2, 1)

(C) (3, 3, -1)

(D) (−3, 1, 1)

Q307. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The sum of the intercepts on the coordinate axes of the plane passing through the point $(-2, -2, 2)$ and containing the line joining the points $(1, -1, 2)$ and (1, 1, 1) is :

(A) 4

(B) -4

(C) -8

(D) 12

Q308. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

If L_1 is the line of intersection of the planes 2x - 2y + 3z - 2 = 0, x - y + z + 1 = 0 and L_2 is the line of intersection of the planes x + 2y - z - 3 = 0, 3x - y + 2z - 1 = 0, then the distance of the origin from the plane, containing the lines L_1 and L_2, is :

(A) $\frac{1}{\sqrt{2}}$

(B) $\frac{1}{4\sqrt{2}}$

(C) $\frac{1}{3\sqrt{2}}$

(D) $\frac{1}{2\sqrt{2}}$

Q309. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

The length of the projection of the line segment joining the points (5, -1, 4) and (4, -1, 3) on the plane, x + y + z = 7 is :

(A) $\sqrt{\frac{2}{3}}$

(B) $\frac{2}{\sqrt{3}}$

(C) $\frac{2}{3}$

(D) $\frac{1}{3}$

Q310. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

A plane bisects the line segment joining the points (1, 2, 3) and $(-3, 4, 5)$ at rigt angles. Then this plane also passes through the point :

(A) $(-3, 2, 1)$

(B) (3, 2, 1)

(C) $(-1, 2, 3)$

(D) $(1, 2, -3)$

Q311. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

A variable plane passes through a fixed point (3,2,1) and meets x, y and z axes at A, B and C respectively. A plane is drawn parallel to yz -plane through A, a second plane is drawn parallel zx-plane through B and a third plane is drawn parallel to xy-plane through C. Then the locus of the point of intersection of these three planes, is :

(A) $\frac{x}{3}+\frac{y}{2}+\frac{z}{1}=1$

(B) x + y + z = 6

(C) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{11}{6}$

(D) $\frac{3}{x}+\frac{2}{y}+\frac{1}{z}=1$

Q312. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

An angle between the plane, x + y + z = 5 and the line of intersection of the planes, 3x + 4y + z − 1 = 0 and 5x + 8y + 2z + 14 =0, is :

(A) $\sin^{-1}\left(\sqrt{3/17}\right)$

(B) $\cos^{-1}\left(\sqrt{3/17}\right)$

(C) $\cos^{-1}\left(3/17\right)$

(D) $\sin^{-1}\left(3/17\right)$

Q313. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If a variable plane, at a distance of 3 units from the origin, intersects the coordinate axes at A, B and C, then the locus of the centroid of ΔABC is :

(A) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=1$

(B) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=3$

(C) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{9}$

(D) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=9$

Q314. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the line, $\frac{x-3}{1}=\frac{y+2}{-1}=\frac{z+\lambda}{-2}$ lies in the plane, 2x−4y+3z=2, then the shortest distance between this line and the line, $\frac{x-1}{12}=\frac{y}{9}=\frac{z}{4}$ is :

(A) 2 (B) 1 (C) 0 (D) 3

Q315. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The coordinates of the foot of the perpendicular from the point (1, −2, 1) on the plane containing the lines, $\frac{x+1}{6}=\frac{y-1}{7}=\frac{z-3}{8}$ and $\frac{x-1}{3}=\frac{y-2}{5}=\frac{z-3}{7}$, is :

(A) (2, −4, 2) (B) (− 1, 2, −1) (C) (0, 0, 0) (D) (1, 1, 1)

Q316. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The line of intersection of the planes $\vec{r}.\left(3\hat{i}-\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(\hat{i}+4\hat{j}-2\hat{k}\right)=2$, is :

(A) $\frac{x-\frac{4}{7}}{-2}=\frac{y}{7}=\frac{z-\frac{5}{7}}{13}$

(B) $\frac{x-\frac{4}{7}}{2}=\frac{y}{-7}=\frac{z+\frac{5}{7}}{13}$

(C) $\frac{x-\frac{6}{13}}{2}=\frac{y-\frac{5}{13}}{-7}=\frac{z}{-13}$

(D) $\frac{x-\frac{6}{13}}{2}=\frac{y-\frac{5}{13}}{7}=\frac{z}{-13}$

Q317. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The distance of the point (1, 3, – 7) from the plane passing through the point (1, –1, – 1), having normal perpendicular to both the lines

$\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$

and

$\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ is :

(A) $\frac{10}{\sqrt{83}}$ (B) $\frac{5}{\sqrt{83}}$ (C) $\frac{10}{\sqrt{74}}$ (D) $\frac{20}{\sqrt{74}}$

Q318. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

If the image of the point P(1, –2, 3) in the plane, 2x + 3y – 4z + 22 = 0 measured parallel to the line,

$\frac{x}{1}=\frac{y}{4}=\frac{z}{5}$ is Q, then PQ is equal to:

(A) $2\sqrt{42}$ (B) $\sqrt{42}$ (C) $6\sqrt{5}$ (D) $3\sqrt{5}$

Q319. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The number of distinct real values of λ for which the lines

$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+3}{\lambda^2}$ and $\frac{x-3}{1}=\frac{y-2}{\lambda^2}=\frac{z-1}{2}$ are coplanar is :

(A) 4 (B) 1 (C) 2 (D) 3

Q320. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The distance of the point (1, – 2, 4) from the plane passing through the point (1, 2, 2) and perpendicular to the planes x – y + 2z = 3 and 2x – 2y + z + 12 = 0, is :

(A) $2\sqrt{2}$ (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Q321. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

If the line, $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z+4}{3}$ lies in the planes, $lx + my - z = 9$, then $l^2 + m^2$ is equal to :

(A) 5 (B) 2 (C) 26 (D) 18

Q322. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along the line $x = y = z$ is :

(A) $\frac{10}{\sqrt{3}}$ (B) $\frac{20}{3}$ (C) $3\sqrt{10}$ (D) $10\sqrt{3}$

Q323. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The equation of the plane containing the line $2x - 5y + z = 3; x + y + 4z = 5$, and parallel to the plane, $x + 3y + 6z = 1$, is :

(A) $x + 3y + 6z = 7$ (B) $2x + 6y + 12z = -13$

(C) $2x + 6y + 12z = 13$ (D) $x + 3y + 6z = -7$

Q324. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The distance of the point $(1, 0, 2)$ from the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 16$, is :

(A) $3\sqrt{21}$ (B) 13 (C) $2\sqrt{14}$ (D) 8

Q325. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$ is the line :

(A) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$ (B) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$

(C) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$ (D) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$

Q326. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

Distance between two parallel planes $2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

Q327. AIEEE 2012 · MCQ (Single Correct Answer)

A equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is :

(A) $x - 2y + 2z - 3 = 0$ (B) $x - 2y + 2z + 1 = 0$

(C) $x - 2y + 2z - 1 = 0$ (D) $x - 2y + 2z + 5 = 0$

Q328. AIEEE 2011 · MCQ (Single Correct Answer)

If the angle between the line $x = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and the plane

$x + 2y + 3z = 4$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then λ equals :

(A) $\frac{3}{2}$ (B) $\frac{2}{5}$ (C) $\frac{5}{3}$ (D) $\frac{2}{3}$

Q329. AIEEE 2010 · MCQ (Single Correct Answer)

Statement-1 : The point $A(3, 1, 6)$ is the mirror image of the point $B(1, 3, 4)$ in the plane $x - y + z = 5$.

Statement-2 : The plane $x - y + z = 5$ bisects the line segment joining $A(3, 1, 6)$ and $B(1, 3, 4)$.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement - 2 is false.

(C) Statement - 1 is false , Statement - 2 is true.

(D) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1.

Q330. AIEEE 2009 · MCQ (Single Correct Answer)

Let the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + \beta = 0$. Then (α, β) equals

(A) $(-6, 7)$ (B) $(5, -15)$ (C) $(-5, 5)$ (D) $(6, -17)$

Q331. AIEEE 2007 · MCQ (Single Correct Answer)

If $(2, 3, 5)$ is one end of a diameter of the sphere $x^2 + y^2 + z^2 - 6x - 12y - 2z + 20 = 0$, then the coordinates of the other end of the diameter are

(A) $(4, 3, 5)$ (B) $(4, 3, -3)$ (C) $(4, 9, -3)$ (D) $(4, -3, 3)$

Q332. AIEEE 2006 · MCQ (Single Correct Answer)

The image of the point $(-1, 3, 4)$ in the plane $x - 2y = 0$ is :

(A) $\left(-\frac{17}{3}, -\frac{19}{3}, 4\right)$ (B) $(15, 11, 4)$

(C) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (D) None of these

Q333. AIEEE 2005 · MCQ (Single Correct Answer)

If the angle θ between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and

the plane $2x - y + \sqrt{\lambda}\, z + 4 = 0$ is such that $\sin\ \theta = \frac{1}{3}$ then value of λ is :

(A) $\frac{5}{3}$ (B) $\frac{-3}{5}$ (C) $\frac{3}{4}$ (D) $\frac{-4}{3}$

Q334. AIEEE 2005 · MCQ (Single Correct Answer)

If the plane $2ax - 3ay + 4az + 6 = 0$ passes through the midpoint of the line joining the centres of the spheres

$x^2 + y^2 + z^2 + 6x - 8y - 2z = 13$ and

$x^2 + y^2 + z^2 - 10x + 4y - 2z = 8$ then a equals :

(A) -1 (B) 1 (C) -2 (D) 2

Q335. AIEEE 2005 · MCQ (Single Correct Answer)

The distance between the line

$\overrightarrow{r} = 2\hat{i} - 2\hat{j} + 3\widehat{k} + \lambda\,(i - j + 4k)$, and the plane

$\overrightarrow{r}.\left(\hat{i} + 5\hat{j} + \widehat{k}\right) = 5$ is

(A) $\frac{10}{9}$ (B) $\frac{10}{3\sqrt{3}}$ (C) $\frac{3}{10}$ (D) $\frac{10}{3}$

Q336. AIEEE 2005 · MCQ (Single Correct Answer)

The plane $x + 2y - z = 4$ cuts the sphere $x^2 + y^2 + z^2 - x + z - 2 = 0$ in a circle of radius

(A) 3 (B) 1 (C) 2 (D) $\sqrt{2}$

Q337. AIEEE 2004 · MCQ (Single Correct Answer)

Distance between two parallel planes

$2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is :

(A) $\frac{9}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{3}{2}$

Q338. AIEEE 2004 · MCQ (Single Correct Answer)

The intersection of the spheres
$x^2 + y^2 + z^2 + 7x - 2y - z = 13$ and
$x^2 + y^2 + z^2 - 3x + 3y + 4z = 8$
is the same as the intersection of one of the sphere and the plane

(A) $2x - y - z = 1$ (B) $x - 2y - z = 1$ (C) $x - y - 2z = 1$ (D) $x - y - z = 1$

Q339. AIEEE 2003 · MCQ (Single Correct Answer)

The shortest distance from the plane $12x + 4y + 3z = 327$ to the sphere

$x^2 + y^2 + z^2 + 4x - 2y - 6z = 155$ is

(A) 39 (B) 26 (C) $11\frac{4}{13}$ (D) 13

Q340. AIEEE 2003 · MCQ (Single Correct Answer)

The radius of the circle in which the sphere

$x^2 + y^2 + z^2 + 2x - 2y - 4z - 19 = 0$ is cut by the plane

$x + 2y + 2z + 7 = 0$ is

(A) 4 (B) 1 (C) 2 (D) 3

Q341. AIEEE 2003 · MCQ (Single Correct Answer)

Two systems of rectangular axes have the same origin. If a plane cuts then at distances a, b, c and a', b', c' from the origin then

(A) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} - \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(B) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2} = 0$

(C) $\frac{1}{a^2} + \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(D) $\frac{1}{a^2} - \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

Q342. AIEEE 2002 · MCQ (Single Correct Answer)

A plane which passes through the point $(3, 2, 0)$ and the line

$\frac{x-4}{1} = \frac{y-7}{5} = \frac{z-4}{4}$ is :

(A) $x - y + z = 1$ (B) $x + y + z = 5$ (C) $x + 2y - z = 1$ (D) $2x - y + z = 5$

Q343. AIEEE 2002 · MCQ (Single Correct Answer)

The $d.r.$ of normal to the plane through $(1, 0, 0), (0, 1, 0)$ which makes an angle $\pi/4$ with plane $x + y = 3$ are :

(A) $1, \sqrt{2}, 1$ (B) $1, 1, \sqrt{2}$ (C) $1, 1, 2$ (D) $\sqrt{2}, 1, 1$

Q344. JEE Main 2023 (Online) 15th April Morning Shift · Numerical

Let the plane P contain the line $2x + y - z - 3 = 0 = 5x - 3y + 4z + 9$ and be parallel to the line $\frac{x+2}{2} = \frac{3-y}{-4} = \frac{z-7}{5}$. Then the distance of the point $\mathrm{A}(8, -1, -19)$ from the plane P measured parallel to the line $\frac{x}{-3} = \frac{y-5}{4} = \frac{2-z}{-12}$ is equal to _____.

Q345. JEE Main 2023 (Online) 13th April Morning Shift · Numerical

Let the image of the point $\left(\frac{5}{3}, \frac{5}{3}, \frac{8}{3}\right)$ in the plane $x - 2y + z - 2 = 0$ be P. If the distance of the point $Q(6, -2, \alpha), \alpha > 0$, from P is 13 , then α is equal to _____.

Q346. JEE Main 2023 (Online) 12th April Morning Shift · Numerical

Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C. If the orthocenter of the triangle ABC is $\left(\alpha, \beta, \frac{6}{7}\right)$, then $98(\alpha + \beta)^2$ is equal to _____.

Q347. JEE Main 2023 (Online) 11th April Evening Shift · Numerical

Let the line $l : x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ meet the plane $P : x + 2y + 3z = 4$ at the point (α, β, γ). If the angle between the line l and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha + 2\beta + 6\gamma$ is equal to _____.

Q348. JEE Main 2023 (Online) 10th April Evening Shift · Numerical

Let the foot of perpendicular from the point $\mathrm{A}(4, 3, 1)$ on the plane $\mathrm{P} : x - y + 2z + 3 = 0$ be N. If $\mathrm{B}(5, \alpha, \beta), \alpha, \beta \in \mathbb{Z}$ is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2 + \beta^2 + \alpha\beta$ is equal to _____.

Q349. JEE Main 2023 (Online) 8th April Evening Shift · Numerical

Let P_1 be the plane $3x - y - 7z = 11$ and P_2 be the plane passing through the points $(2, -1, 0), (2, 0, -1)$, and $(5, 1, 1)$. If the foot of the perpendicular drawn from the point $(7, 4, -1)$ on the line of intersection of the planes P_1 and P_2 is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____.

Q350. JEE Main 2023 (Online) 8th April Morning Shift · Numerical

Let λ_1, λ_2 be the values of λ for which the points $\left(\frac{5}{2}, 1, \lambda\right)$ and $(-2, 0, 1)$ are at equal distance from the plane $2x + 3y - 6z + 7 = 0$. If $\lambda_1 > \lambda_2$, then the distance of the point $(\lambda_1 - \lambda_2, \lambda_2, \lambda_1)$ from the line $\frac{x-5}{1} = \frac{y-1}{2} = \frac{z+7}{2}$ is _____.

Q351. JEE Main 2023 (Online) 6th April Morning Shift · Numerical

Let the image of the point $\mathrm{P}(1, 2, 3)$ in the plane $2x - y + z = 9$ be Q. If the coordinates of the point R are $(6, 10, 7)$, then the square of the area of the triangle PQR is _____.

Q352. JEE Main 2023 (Online) 1st February Evening Shift · Numerical

The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $\mathrm{A}(-3, -6, 1)$ and $\mathrm{B}(2, 4, -3)$ divides the line segment AB internally in the ratio $\mathrm{k} : 1$. If $\mathrm{a}, \mathrm{b}, \mathrm{c}(|\mathrm{a}|, |\mathrm{b}|, |\mathrm{c}|$ are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $|\mathrm{a} + \mathrm{b} + \mathrm{c}|$ is equal to _____.

Q353. JEE Main 2023 (Online) 1st February Evening Shift · **Numerical**

Let $\alpha x + \beta y + \gamma z = 1$ be the equation of a plane passing through the point $(3, -2, 5)$ and perpendicular to the line joining the points $(1, 2, 3)$ and $(-2, 3, 5)$. Then the value of $\alpha\beta\gamma$ is equal to _____.

Q354. JEE Main 2023 (Online) 31st January Morning Shift · **Numerical**

Let the line $L : \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P. Let the point Q be the foot of perpendicular from the point $R(1, -1, -3)$ on the line L. If α is the area of triangle PQR, then α^2 is equal to _____.

Q355. JEE Main 2023 (Online) 31st January Morning Shift · **Numerical**

Let θ be the angle between the planes $P_1 : \vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 9$ and $P_2 : \vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2. If α is the angle between L and P_2, then $\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$ is equal to _____.

Q356. JEE Main 2023 (Online) 30th January Morning Shift · **Numerical**

If the equation of the plane passing through the point $(1, 1, 2)$ and perpendicular to the line $x - 3y + 2z - 1 = 0 = 4x - y + z$ is $\mathrm{A}x + \mathrm{B}y + \mathrm{C}z = 1$, then $140(\mathrm{C} - \mathrm{B} + \mathrm{A})$ is equal to _____.

Q357. JEE Main 2023 (Online) 30th January Morning Shift · **Numerical**

If $\lambda_1 < \lambda_2$ are two values of λ such that the angle between the planes $P_1 : \vec{r}(3\hat{i} - 5\hat{j} + \hat{k}) = 7$ and $P_2 : \vec{r} \cdot (\lambda\hat{i} + \hat{j} - 3\hat{k}) = 9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is _____.

Q358. JEE Main 2023 (Online) 29th January Morning Shift · **Numerical**

Let the equation of the plane P containing the line $x + 10 = \frac{8-y}{2} = z$ be $ax + by + 3z = 2(a + b)$ and the distance of the plane P from the point (1, 27, 7) be c. Then $a^2 + b^2 + c^2$ is equal to _____.

Q359. JEE Main 2023 (Online) 25th January Morning Shift · **Numerical**

Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point (a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____.

Q360. JEE Main 2022 (Online) 29th July Morning Shift · **Numerical**

Let a line with direction ratios $a, -4a, -7$ be perpendicular to the lines with direction ratios $3, -1, 2b$ and $b, a, -2$. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x - y + z = 0$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____.

Q361. JEE Main 2022 (Online) 27th July Morning Shift · Numerical

Let the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ and $4ax-y+5z-7a=0=2x-5y-z-3, a\in\mathbb{R}$ at the point $P(\alpha,\beta,\gamma)$. Then the value of $\alpha+\beta+\gamma$ equals _____.

Q362. JEE Main 2022 (Online) 26th July Evening Shift · Numerical

The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+a\hat{j}-\hat{k})$ and $\vec{r}=(\hat{i}+\hat{j})+\mu(-\hat{i}+\hat{j}-a\hat{k})$ from the point $(2,1,4)$ is $\sqrt{3}$, is _____.

Q363. JEE Main 2022 (Online) 26th July Evening Shift · Numerical

The plane passing through the line $L: lx-y+3(1-l)z=1, x+2y-z=2$ and perpendicular to the plane $3x+2y+z=6$ is $3x-8y+7z=4$. If θ is the acute angle between the line L and the y-axis, then $415\cos^2\theta$ is equal to _____.

Q364. JEE Main 2022 (Online) 25th July Morning Shift · Numerical

The line of shortest distance between the lines $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ and $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane $\mathrm{P}: \mathrm{a}x-y-z=0, (a>0)$. If the image of the point $(1,1,-5)$ in the plane P is (α,β,γ), then $\alpha+\beta-\gamma$ is equal to _____.

Q365. JEE Main 2022 (Online) 29th June Morning Shift · Numerical

Let d be the distance between the foot of perpendiculars of the points P(1, 2, -1) and Q(2, -1, 3) on the plane $-$x + y + z = 1. Then d^2 is equal to _____.

Q366. JEE Main 2022 (Online) 29th June Morning Shift · Numerical

Let $P_1: \vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4$ be a plane. Let P_2 be another plane which passes through the points (2, -3, 2), (2, -2, -3) and (1, -4, 2). If the direction ratios of the line of intersection of P_1 and P_2 be 16, α, β, then the value of $\alpha+\beta$ is equal to _____.

Q367. JEE Main 2022 (Online) 27th June Morning Shift · Numerical

Let the mirror image of the point (a, b, c) with respect to the plane 3x $-$ 4y + 12z + 19 = 0 be (a $-$ 6, β, γ). If a + b + c = 5, then $7\beta-9\gamma$ is equal to _____.

Q368. JEE Main 2022 (Online) 25th June Morning Shift · **Numerical**

Let the lines

$L_1 : \overrightarrow{r} = \lambda\left(\hat{i} + 2\hat{j} + 3\hat{k}\right),\ \lambda \in R$

$L_2 : \overrightarrow{r} = \left(\hat{i} + 3\hat{j} + \hat{k}\right) + \mu\left(\hat{i} + \hat{j} + 5\hat{k}\right);\ \mu \in R,$

intersect at the point S. If a plane ax + by $-$ z + d = 0 passes through S and is parallel to both the lines L_1 and L_2, then the value of a + b + d is equal to _____.

Q369. JEE Main 2021 (Online) 31st August Evening Shift · **Numerical**

Suppose, the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. Then $(\alpha + \beta)$ is equal to _____.

Q370. JEE Main 2021 (Online) 31st August Morning Shift · **Numerical**

The square of the distance of the point of intersection

of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is _____.

Q371. JEE Main 2021 (Online) 27th August Evening Shift · **Numerical**

Let S be the mirror image of the point Q(1, 3, 4) with respect to the plane 2x $-$ y + z + 3 = 0 and let R(3, 5, γ) be a point of this plane. Then the square of the length of the line segment SR is _____.

Q372. JEE Main 2021 (Online) 26th August Evening Shift · **Numerical**

Let Q be the foot of the perpendicular from the point P(7, -2, 13) on the plane containing the lines $\frac{x+1}{6} = \frac{y-1}{7} = \frac{z-3}{8}$ and $\frac{x-1}{3} = \frac{y-2}{5} = \frac{z-3}{7}$. Then $(PQ)^2$, is equal to _____.

Q373. JEE Main 2021 (Online) 26th August Morning Shift · **Numerical**

Let the line L be the projection of the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ in the plane x $-$ 2y $-$ z = 3. If d is the distance of the point (0, 0, 6) from L, then d^2 is equal to _____.

Q374. JEE Main 2021 (Online) 27th July Evening Shift · **Numerical**

The distance of the point P(3, 4, 4) from the point of intersection of the line joining the points Q(3, -4, -5) and R(2, -3, 1) and the plane 2x + y + z = 7, is equal to _____.

Q375. JEE Main 2021 (Online) 27th July Morning Shift · Numerical

Let a plane P pass through the point (3, 7, -7) and contain the line, $\frac{x-2}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$. If distance of the plane P from the origin is d, then d^2 is equal to _____.

Q376. JEE Main 2021 (Online) 25th July Evening Shift · Numerical

If the lines $\frac{x-k}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and
$\frac{x+1}{3} = \frac{y+2}{2} = \frac{z+3}{1}$ are co-planar, then the value of k is _____.

Q377. JEE Main 2021 (Online) 20th July Morning Shift · Numerical

Let P be a plane passing through the points (1, 0, 1), (1, -2, 1) and (0, 1, -2). Let a vector $\overrightarrow{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ be such that $\overrightarrow{a}$ is parallel to the plane P, perpendicular to $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $\overrightarrow{a}.\ (\hat{i} + \hat{j} + 2\hat{k}) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____.

Q378. JEE Main 2021 (Online) 18th March Evening Shift · Numerical

Let the mirror image of the point (1, 3, a) with respect to the plane $\overrightarrow{r}.\left(2\hat{i} - \hat{j} + \hat{k}\right) - b = 0$ be (-3, 5, 2). Then, the value of | a + b | is equal to _____.

Q379. JEE Main 2021 (Online) 18th March Evening Shift · Numerical

Let P be a plane containing the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+5}{2}$ and parallel to the line $\frac{x-1}{4} = \frac{y-2}{-3} = \frac{z+5}{7}$. If the point (1, -1, α) lies on the plane P, then the value of $|5\alpha|$ is equal to _____.

Q380. JEE Main 2021 (Online) 18th March Morning Shift · Numerical

Let the plane ax + by + cz + d = 0 bisect the line joining the points (4, -3, 1) and (2, 3, -5) at the right angles. If a, b, c, d are integers, then the
minimum value of ($a^2 + b^2 + c^2 + d^2$) is _____.

Q381. JEE Main 2021 (Online) 18th March Morning Shift · Numerical

The equation of the planes parallel to the plane x $-$ 2y + 2z $-$ 3 = 0 which are at unit distance from the point (1, 2, 3) is ax + by + cz + d = 0. If (b $-$ d) = k(c $-$ a), then the positive value of k is :

Q382. JEE Main 2021 (Online) 17th March Evening Shift · Numerical

Let P be an arbitrary point having sum of the squares of the distances from the planes x + y + z = 0, lx $-$ nz = 0 and x $-$ 2y + z = 0, equal to 9. If the locus of the point P is $x^2 + y^2 + z^2 = 9$, then the value of l $-$ n is equal to _____.

Q383. JEE Main 2021 (Online) 17th March Morning Shift · **Numerical**

If the equation of the plane passing through the line of intersection of the planes $2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is _____.

Q384. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line, $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of |m| is equal to _____.

Q385. JEE Main 2021 (Online) 26th February Morning Shift · **Numerical**

Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the points $(-2, -21, 29)$ and $(-1, -16, 23)$, then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to _____.

Q386. JEE Main 2020 (Online) 4th September Morning Slot · **Numerical**

If the equation of a plane P, passing through the intersection of the planes,
$x + 4y - z + 7 = 0$ and $3x + y + 5z = 8$ is $ax + by + 6z = 15$ for some $a, b \in R$, then the distance of the point (3, 2, -1) from the plane P is.....

Q387. JEE Main 2020 (Online) 3rd September Evening Slot · **Numerical**

Let a plane P contain two lines
$\overrightarrow{r} = \hat{i} + \lambda\left(\hat{i} + \hat{j}\right), \lambda \in R$ and
$\overrightarrow{r} = -\hat{j} + \mu\left(\hat{j} - \widehat{k}\right), \mu \in R$
If $Q(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the point M(1, 0, 1) to P, then $3(\alpha + \beta + \gamma)$ equals _____.

Q388. JEE Main 2020 (Online) 9th January Evening Slot · **Numerical**

If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane

containing the lines $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3}$

and $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$ $(\lambda \in R)$

is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **C**	3. **C**	4. **B**	5. **C**
6. **C**	7. **C**	8. **C**	9. **B**	10. **A**
11. **B**	12. **B**	13. **A**	14. **C**	15. **C**
16. **B**	17. **C**	18. **A**	19. **A**	20. **C**
21. **B**	22. **D**	23. **A**	24. **A**	25. **D**
26. **A**	27. **A**	28. **B**	29. **B**	30. **B**
31. **A**	32. **C**	33. **D**	34. **D**	35. **D**
36. **C**	37. **C**	38. **D**	39. **C**	40. **A**
41. **A**	42. **A**	43. **A**	44. **A**	45. **D**
46. **B**	47. **A**	48. **C**	49. **A**	50. **C**
51. **B**	52. **D**	53. **C**	54. **D**	55. **D**
56. **C**	57. **C**	58. **C**	59. **A**	60. **B**
61. **B**	62. **A**	63. **B**	64. **A**	65. **B**
66. **B**	67. **D**	68. **D**	69. **D**	70. **A**
71. **A**	72. **B**	73. **A**	74. **D**	75. **B**
76. **C**	77. **A**	78. **A**	79. **C**	80. **C**
81. **D**	82. **C**	83. **C**	84. **D**	85. **D**
86. **A**	87. **B**	88. **C**	89. **D**	90. **A**
91. **C**	92. **D**	93. **A**	94. **C**	95. **C**
96. **D**	97. **B**	98. **B**	99. **B**	100. **A**
101. **A**	102. **A**	103. **A**	104. **B**	105. **A**
106. **C**	107. **A**	108. **A**	109. **A**	110. **A**
111. **A**	112. **C**	113. **D**	114. **A**	115. **B**
116. **A**	117. **C**	118. **B**	119. **D**	120. **C**
121. **A**	122. **D**	123. **D**	124. **A**	125. **C**
126. **B**	127. **C**	128. **C**	129. **C**	130. **C**
131. **A**	132. **B**	133. **B**	134. **C**	135. **A**
136. **B**	137. **C**	138. **A**	139. **B**	140. **C**
141. **B**	142. **D**	143. **A**	144. **D**	

Numerical

145. **4**	146. **162**	147. **170**	148. **6**	149. **3**
150. **56**	151. **957**	152. **216**	153. **62**	154. **11**
155. **43**	156. **13**	157. **25**	158. **6**	159. **21**
160. **22**	161. **12**	162. **196**	163. **16**	164. **9**
165. **65**	166. **108**	167. **5**	168. **18**	169. **158**
170. **9**	171. **18**	172. **384**	173. **14**	174. **450**
175. **153**	176. **6**	177. **125**	178. **51**	179. **84**
180. **2**	181. **6**	182. **44**	183. **1**	184. **8**
185. **4**				

Out of syllabus

186. **C**	187. **D**	188. **A**	189. **D**	190. **C**
191. **A**	192. **D**	193. **B**	194. **A**	195. **A**
196. **D**	197. **C**	198. **B**	199. **D**	200. **C**
201. **A**	202. **B**	203. **B**	204. **B**	205. **C**
206. **A**	207. **D**	208. **B**	209. **A**	210. **A**
211. **D**	212. **C**	213. **A**	214. **A**	215. **A**
216. **D**	217. **B**	218. **D**	219. **B**	220. **B**
221. **D**	222. **B**	223. **D**	224. **B**	225. **B**
226. **B**	227. **D**	228. **C**	229. **C**	230. **C**
231. **B**	232. **B**	233. **C**	234. **D**	235. **B**
236. **A**	237. **B**	238. **A**	239. **C**	240. **B**
241. **A**	242. **C**	243. **B**	244. **C**	245. **B**
246. **D**	247. **A**	248. **A**	249. **D**	250. **D**
251. **C**	252. **B**	253. **D**	254. **D**	255. **C**
256. **B**	257. **D**	258. **B**	259. **B**	260. **A**
261. **A**	262. **D**	263. **C**	264. **A**	265. **C**
266. **C**	267. **D**	268. **B**	269. **D**	270. **D**
271. **C**	272. **D**	273. **B**	274. **B**	275. **D**
276. **D**	277. **C**	278. **B**	279. **A**	280. **B**
281. **D**	282. **D**	283. **C**	284. **A**	285. **B**
286. **C**	287. **C**	288. **A**	289. **C**	290. **C**
291. **C**	292. **D**	293. **D**	294. **B**	295. **C**
296. **B**	297. **B**	298. **C**	299. **C**	300. **C**

301. **C**	302. **B**	303. **C**	304. **A**	305. **A**
306. **B**	307. **B**	308. **C**	309. **A**	310. **A**
311. **D**	312. **A**	313. **A**	314. **C**	315. **C**
316. **C**	317. **A**	318. **A**	319. **D**	320. **A**
321. **B**	322. **D**	323. **A**	324. **B**	325. **C**
326. **C**	327. **A**	328. **D**	329. **A**	330. **A**
331. **C**	332. **D**	333. **A**	334. **C**	335. **B**
336. **B**	337. **C**	338. **A**	339. **D**	340. **D**
341. **A**	342. **A**	343. **B**	344. **26**	345. **15**
346. **288**	347. **11**	348. **7**	349. **11**	350. **9**
351. **594**	352. **10**	353. **6**	354. **180**	355. **9**
356. **15**	357. **315**	358. **355**	359. **9**	360. **10**
361. **12**	362. **2**	363. **125**	364. **3**	365. **26**
366. **28**	367. **137**	368. **5**	369. **7**	370. **61**
371. **72**	372. **96**	373. **26**	374. **7**	375. **3**
376. **1**	377. **81**	378. **1**	379. **38**	380. **28**
381. **4**	382. **0**	383. **4**	384. **2**	385. **8**
386. **3**	387. **5**	388. **3**		